

GoFish: A Next-Generation ToolKit for
Modeling Migratory Fish and Quantifying
Sediment-Bound Contaminant Risk in Estuaries

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Contents

Preface

This library provides a comprehensive, modular framework for developing and documenting agent-based models (ABMs) that simulate the movement, behavior, and environmental interactions of migratory fish in coastal aquatic systems. It is designed to support the standardization and implementation of ABMs in fisheries management, enabling researchers and practitioners to address complex environmental questions and evaluate remediation or restoration scenarios.

Motivation

The goal of this resource is to support students, researchers, and decision-makers by making agent-based modeling of migratory fish more accessible, reproducible, and applicable to real-world fisheries and habitat management challenges. By providing a standardized modular framework for key behavioral processes, this library promotes consistency, transparency, and credibility in ecological forecasting and decision support tools. It also establishes a foundation for critical conversations about the behaviors and functions represented in modeling, while supporting the empirical quantification of ecological relationships that influence movement, survival, and habitat use of migratory fish.

Application Context

This library was originally developed in support of research on the influence of tidal behavior and contaminant exposure on anadromous fish in the Penobscot River Estuary. However, its modular design allows for application to other estuarine and coastal systems where fish respond to environmental changes (i.e., salinity, velocity, and pollutants).

How to Use This Library

Each function or behavior in this library can be combined with others to build a complete agent-based model for anadromous fish. These functions are designed

to be adaptable, and easily configured for different species, life stages, or site-specific conditions.

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*For questions, feedback, guidance on implementation, or **interest in adding to the library**, please contact **Vanessa Quintana** at **ma-han.vanessa98@gmail.com**.

Chapter 1

Introduction

Anadromous fish species such as river herring, striped bass, salmon, and sturgeon navigate coastal and estuarine systems that are increasingly affected by human activity, climate change, and legacy contaminants. Modeling their movement and behavior at fine spatial and temporal scales requires tools that can integrate physiological stressors, environmental variability, and behavior-based decision-making.

Agent-based models are among the most powerful tools available for ecological forecasting and fisheries management, but they are also among the most complex. Their structure and computational demands can make them difficult to apply in practical management settings. Many biologists and ecologists who hold deep, species-specific expertise often have limited training in advanced programming or systems modeling.

As a result, traditional approaches to simulating anadromous fish frequently oversimplify or exclude key biological functions and environmental interactions such as salinity exposure and contaminant exposure. Existing models also lack standardized representations of these behaviors and interactions, limiting the applicability of results and reducing their usefulness for applied management. This function library was developed to address these limitations by offering modular, empirically grounded components designed for use in modeling migratory fish. Each function is clearly documented and can be applied independently, allowing for transparent testing, modification, and reuse across a wide range of ecological scenarios and sites.

1.1 Structure

Each chapter in this library corresponds to a major behavior, physiological function, or foundational component relevant to migratory fish, beginning with:

- **Co-Development of Agent-Based Models**
- **Calculate Metabolism**
- **Salinity Exposure**
- **Temperature Exposure**
- **Contaminant Exposure**
- **Digestion**
- **Filter Feeding**
- **Lipid Catabolism**
- **Landward Migration**
- **Seaward Migration**
- **Schooling**
- **Selective Tidal Stream Transport**
- **Predator-prey interactions**

The final chapters provide guidance on how to integrate multiple functions into a complete agent-based model, demonstrating how these components work together to simulate fish behavior in dynamic coastal and estuarine systems.

Within each chapter, each function or behavior is documented using the ODD protocol (???). The ODD (Overview, Design concepts, Details) protocol is a standardized framework for describing agent-based models. It promotes transparency in model development and ensures consistency across implementations, especially when integrating multiple behavioral or ecological functions.

- **Overview** provides the purpose of the model component, identifies the entities involved (e.g., fish agents, environmental patches), and outlines the general processes.
- **Design** concepts describe the key theoretical underpinnings such as emergence, adaptation, objectives, sensing, stochasticity, and interaction.
- **Details** specify initialization steps, input data requirements, and the rules or submodels that govern behavior.

By following the ODD protocol, this library ensures that each function is self-contained, interpretable, and ready for adaptation to a wide range of species, sites, or management scenarios.

Chapter 2

Agent-Based Models

Agent-based models (ABMs) simulate the behavior of individual organisms—called “agents”—and how their interactions with each other and with their environment create larger-scale patterns. Each agent follows a set of simple rules based on physiology, behavior, or environmental cues. When many agents follow these rules simultaneously, complex system-level outcomes emerge, such as migration pathways, schooling patterns, habitat bottlenecks, or exposure hotspots. ABMs are especially useful for ecological systems because they allow researchers and communities to explore how individual decisions scale up to influence population dynamics, habitat use, and risk in changing environments.

ABMs are designed to capture processes that depend on **behavior, timing, environmental variability**, and **individual differences**. They allow modelers to explore how organisms respond to their surroundings, make movement decisions, and interact with other individuals or species. Because agents experience the environment at fine spatial and temporal scales, ABMs can reveal patterns that occur only when many individuals interact across a dynamic landscape.

2.0.1 What ABMs Can Help Us Understand

ABMs can answer questions related to:

- How individual movement decisions accumulate into migration routes or staging locations
- When and where organisms are most exposed to environmental risks
- How behaviors such as foraging, schooling, predator avoidance, resting, or spawning influence exposure or survival
- How variation in traits (size, energy, age, or species) contributes to different outcomes

- How environmental conditions shape habitat use, movement choices, or interactions between species
- How system-level patterns emerge from local behaviors and simple rules
- How alternative conditions or restoration actions might alter movement, habitat use, or exposure

These models are especially useful when management questions depend on **patterns created by individual behavior**, rather than population totals alone.

2.0.2 What ABMs Are Not Designed to Do

ABMs cannot provide or replace:

- Exact estimates of population abundance
- Precise contaminant concentrations inside individual organisms
- Direct measurements of chemical or physiological processes
- Predictions of exact numbers of predation or spawning events
- Large-scale demographic projections without additional modeling components
- Replacements for field sampling, monitoring programs, or laboratory measurements

Instead, ABMs offer **process-based insight**, showing how behavior and environmental conditions interact to create observable patterns.

2.0.3 Emergent Behaviors

Emergent behavior refers to patterns that arise when many individual agents follow their own rules, creating outcomes that cannot be predicted from observing a single fish on its own. These outcomes develop only when many individuals interact with each other and respond to their surrounding environment at the same time. Many ecological processes such as migration timing, habitat bottlenecks, exposure hotspots, schooling dynamics, predator and prey encounters, and shifts in group structure are emergent rather than linear. They form through small, individual decisions that accumulate into larger and sometimes unexpected system-level patterns.

By visualizing and simulating emergence, ABMs make it possible to see how these individual behaviors scale up to influence population level outcomes and habitat use patterns. This helps identify the specific conditions, behaviors, and locations that shape system-wide responses. For management, this is valuable because it shows how small-scale decisions made by individual fish can contribute to broader patterns that influence risk, habitat quality, restoration effectiveness, management actions, and ecological resilience. Understanding emer-

gence allows managers and communities to explore why certain patterns occur, identify potential leverage points, and anticipate how changes in environmental conditions may alter system behavior in the future.

Chapter 3

Co-Development of ABMs: Penobscot Estuary Community Workshop August 6, 2025

This page documents how a participatory workshop informed the functions within this coding library, the representation of physiological processes, behaviors of migratory fish, and their interactions within an estuary. The purpose of this page is to showcase the value added by co-development and increase transparency about the development of this coding library.

The Penobscot Estuary Community Modeling Workshop was designed from the beginning as a co-development space for an agent based model to address mercury toxicity dynamics of anadromous fish within the estuary. The day invited Tribal collaborators, community members, managers, and researchers to shape the modeling objectives, including which species, processes, behaviors, and outputs the model should represent, and how those outputs should be represented in relation to contaminant exposure in the Penobscot River Estuary.

3.1 Core Co-Development Themes

Several themes guided the design of the workshop and the way it connects to model development.

- **Shared problem framing.** The workshop centered a shared concern about mercury contamination in the Penobscot River and its effects on sea-run fish, subsistence access, and recreation use, rather than starting

from a predefined modeling objective.

- **Attention to both internal and external processes.** Breakout discussions were split into two rounds, one for “internal” or biological functions of fish such as osmoregulation, thermoregulation, bioaccumulation, and resting, and the second round focused on “external processes” or external interactions with intra- or interspecific factors and environmental responses such as swimming, predation, foraging, and spawning. This structure maps directly onto modules detailed within this library.
- **Participant driven priorities.** Participants were asked which species were most important to understand for management of the system or species that should be prioritized based on community interest, which functions and interactions they believed to be relevant for understanding mercury toxicity in migratory fish, and how those behaviors should be represented.
- **Outputs that answer real questions.** Each participant had the opportunity to propose a question or hypothesis to test using the developed functions and select preferred outputs from the functions such as migration paths, exposure metrics, or habitat use. These wish-list requests now correspond to specific model output routines and visualizations.
- **Accessible language and shared vocabulary.** A glossary of terms translated physiological, hydrodynamic, and modeling concepts into accessible language. This glossary can be found at the end of this document and is intended to support a range of readers.
- **Iterative relationships, not a single event.** The workshop was framed as one step in an ongoing partnership. Feedback forms, follow-up contact options, and explicit invitations for later review are built into the process.

3.2 Workshop Development

3.2.1 Pre-Workshop Collaboration Steps

Planning the workshop involved months of coordination with partners, including the Penobscot Nation, faculty and staff at the University of Maine, and federal collaborators. The planning process itself was a first stage of co-development.

Key steps included:

- **Clarifying objectives with partners.** Early conversations identified contamination in the Penobscot River, migratory fish, and access to safe subsistence resources as the shared focus. Partners helped guide agenda development, identify potential participants of interest, and identify community outputs of interest to support Tribal management. Their suggestions were used when conducting outreach.

- **Aligning model scope with community priorities.** The agenda was constructed so that introductory talks on Penobscot Nation natural resource management and contaminants of interest, an overview of estuary dynamics (connection of fish + physical processes + contamination), and an introduction to agent based modeling were followed by interactive surveys and break-out discussions. This allowed participants to identify anadromous species, behaviors, and processes of interest, which then feed directly into the model design.
- **Preparing interactive polls and prompts.** A Mentimeter survey was developed to help participants identify priority species, important environmental drivers, and fish behaviors that shape exposure. These questions now map to the list of behaviors and parameters implemented in the model.

Question Example

- **Designing materials for many kinds of learners.** Each participant received a notepad and workshop folder, which included a plain-language glossary, a participant agenda, a hypothesis and output worksheet, and a feedback form. These materials supported participants who learn best through reading, writing, discussion, or visual exploration. Break-out discussions were accompanied by visual demonstrations of function demos, allowing participants to directly address draft model functions for each behavior.

Find examples below.

Note: these are presentations of functions before participant feedback. Updated functions in this library reflect changes informed by the workshop.

3.2.2 Identifying Participants of Interest

Participant selection was guided by the goal of gathering diverse knowledge about the river and its fish.

- **Tribal Collaborators.** Representatives from the Penobscot Nation were invited to collaborate because the Penobscot River Estuary is the Ancestral Homeland of the Penobscot Nation and other Wabanaki Peoples. It was necessary to prioritize their concerns and support their management. We wanted to give them a platform to share their work and management perspectives on migratory fish and contamination. Indigenous people have sovereign rights to migratory species, which have been deeply impacted by contamination in the estuary.
- **Community Experts.** The workshop brought in local managers, fishermen, researchers, engineers, and students whose work focuses on estuarine dynamics, fisheries, remediation, and environmental management in the Penobscot, or who focus on anadromous fish.

3.2.3 Enticing Participants to Attend

The invitations and overview materials emphasized that this was a hands-on, collaborative event where participant input would shape real model structure and outputs—not a passive lecture.

Copy of Invitation

Strategies included:

- **Clear framing.** The overview highlighted how tidal dynamics, fish behavior, and mercury contamination intersect in the Penobscot River, and described specific opportunities for participants to contribute knowledge during breakout sessions and wish-list activities. The invitation also emphasized that participants did not need any modeling experience—only their expertise and perspectives.
- **Concrete outcomes.** Participants could request a specific model output or test a hypothesis relevant to their interests, with results attempted to be shared after the workshop (as feasible within model limitations).
- **Relationship building.** The agenda previewed time for partnership discussions, highlighting opportunities to support Tribal priorities and future collaboration on management and research questions.

3.2.4 Selecting the Venue

The workshop was held at the Innovation Media Research Center at the University of Maine, a space that could support immersive presentations and movement between small and large group activities.

The venue allowed for:

- **A theater-style room with a large screen.** The introductory talks and interactive survey used a dark background, low lighting, and a large screen to create an immersive experience similar to an intimate concert hall, while still leaving space for questions and discussion.
- **A separate interactive space.** Round tables, easel pads, printed handouts, and discussion areas were set up in a second room for the world-café breakout sessions, allowing participants to interact and face one another in a conversation-style setting as they discussed each function.

3.3 Facilitating the Workshop

3.3.1 Agenda Activities

The agenda combined presentations, interactive polls, world-café discussions, and reflection.

Key components included:

- **Welcome, blessing, and introductions.** The day opened with a welcome and words from the Penobscot Nation, followed by introductions and an icebreaker activity asking participants why they attended and what they hoped to gain.
- **Map-based activities.** Participants could pin locations on a system-scale map, connecting their lived experience and knowledge to specific reaches of the river and estuary. This place-based interaction showcased lived knowledge and gave participants autonomy over the spatial representation of the system.
- **Penobscot Nation history and river overview.** A management representative of the Penobscot Nation, Dan Kircheis, provided management context for contaminants and migratory fish, grounding the model in current work conducted within the system.
- **Modeling overview and agent based model presentation.** Modeling presentations introduced the Penobscot system, key hydrodynamic and contaminant processes, and the rationale for an agent based approach. Examples of emergent behaviors and validation were shown on a large screen to make model dynamics visible and engaging.
- **Mentimeter priority setting.** Participants completed an interactive survey about priority species, important environmental drivers, and key behaviors. These responses were summarized live and used to frame the breakout sessions.
- **World café on internal fish processes.** Small rotating groups discussed processes such as osmoregulation, thermoregulation, bioaccumulation, and resting. Discussion leaders presented demos and recorded notes, which later informed how these processes are parameterized and sequenced within this library.
- **World café on external behaviors and conditions.** A second set of rotations focused on external functions such as foraging, migration, predation, and spawning. These conversations helped identify how fish interact with their environment and where and when the model should track exposure risk.
- **Wish list outputs.** Participants wrote down specific outputs they wanted from the model and the reasoning behind each request, including migration timing, exposure patterns, energy budgets, and habitat use. The goal is to incorporate as many of these outputs as possible to support subsequent management and research.

Hypothesis Form

- **Model demo.** A short demonstration of the model interface supported visual learners and connected the function modules to emergent behaviors.

- **Partnership and next steps discussion.** The workshop closed with a short session on building partnerships and identifying community projects where the model could support ongoing work. This included relationship-building between participants and completion of a follow-up survey.

Participant Survey

- **Facilitation.** Facilitation roles were shared across a team of organizers and volunteers who kept time, prompted discussion, recorded notes, and supported logistics, which allowed the lead modeler to focus on listening and translating insights into model design.

3.4 After the Workshop

3.4.1 Incorporating Feedback

Immediately following the workshop, participants were invited to fill out feedback forms on what they found most useful, what felt unclear or missing, and how future workshops or model development activities could be improved.

Feedback is being used to:

- Refine explanations of agent based modeling and estuarine processes in the model interface and supporting documents, using more examples, more visuals, and clearer connections between field observations and model rules.
- Adjust the balance between presentation time and discussion time.
- Clarify how participant input has been integrated, so that people can see their contributions in the model structure and outputs rather than only in summary notes.

3.4.2 Post-Workshop Follow-Up

Several tools support ongoing co-development after the workshop day itself. This work is still ongoing as products are completed and co-development of final materials continues.

- **Hypothesis and output forms.** The participant hypothesis and output form captured individual questions, preferred outputs, and desired scenarios. These forms now guide scenario design and output routines in the model (as much as feasible within model limitations and scope), and follow-up emails will share results with those who opted in.
- **Ongoing communication.** Participants who requested follow-up are being contacted with model updates, refinements, and opportunities to review model structure.
- **Follow-up survey.** The feedback form includes an option to receive a later survey on model results, which will allow participants to review

outputs, comment on whether they find them credible and useful, and suggest further refinements.

- **Tech Transfer Webinar.** A planned webinar will support technology transfer of the completed model and products to workshop participants, allowing them to see how their input made it into the final product.
- **Integration into model documentation.** This co-development page, the glossary, and workshop-derived figures will be hosted alongside the developed functions to make the participatory process visible to anyone using or reviewing the simulations.

3.4.3 Acknowledgements

The workshop and this co developed model were made possible by the time, knowledge, and care shared by workshop participants and organizers.

3.4.3.1 Participants

Ernie Atkinson
 Andrea Casey
 Devon Gleason
 Heather Hamlin
 John Hildebrand
 Dan Kircheis
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Chapter 4

Metabolism

4.1 Overview

Metabolism determines the baseline and dynamic energy demand required for fish to maintain physiological function, support movement, regulate ions, digest food, and sustain long-term migration. Within P-MEM, metabolic rate is computed as a function of body size, temperature, and age, following established allometric scaling relationships in ectothermic organisms, particularly fishes (e.g., (???)). Temperature sensitivity is represented using a Q10 formulation consistent with ectothermic metabolic responses (?), while age-dependent scaling reflects ontogenetic changes in metabolic intensity (??). Metabolic efficiency, based on divergence from an agent’s optimal temperature, is then used to scale energy allocation across key internal processes including swimming, osmoregulation, and lipid catabolism.

4.2 Purpose

To simulate energetically realistic metabolic costs for migratory fishes in coastal and estuarine environments by accounting for size scaling, temperature dependence, age effects, and dynamic energy partitioning across physiological processes.

4.3 Entities, State Variables, and Scales

4.3.1 Spatial and Temporal Scales

- **Spatial Unit:** Patch (3 m x 3 m resolution)
- **Temporal Unit:** 5-minute time steps (`tick`)

4.3.2 Global Variables

Global Variable	Definition
Q10	Temperature coefficient used for metabolic scaling.
base-constant	Global constant for baseline metabolic scaling (0.1).
age-sensitivity	Scalar controlling how metabolism changes with age (0.5).

4.3.3 Patch Variables

Variable Name	Definition
temperature	The water temperature of the current patch used to calculate temperature dependent metabolic rate.

4.3.4 Agent Variables

Variable Name	Definition
size	Body mass or size index used for allometric scaling.
age-num	Age used in scaling metabolic rate.
optimal-temperature	Preferred temperature at which baseline metabolism is defined.
max-survival-temp	Maximum temperature at which the agent can survive and function.
Met-base	Baseline metabolic constant calculated during initialization.
metabolism-rate	Metabolic rate for the current tick.
total-metabolism	Accumulated metabolism over time.
swim-efficiency	Efficiency applied to movement when metabolic rate is lower or higher than baseline.

Variable Name	Definition
osmoregulation-efficiency	Efficiency applied to osmoregulation when metabolic rate is lower or higher than baseline.
lipid-catabolism-efficiency	Efficiency applied to lipid catabolism when metabolic rate is lower or higher than baseline.
swim-base	Baseline allocation of metabolism toward movement.
E-base	Base metabolic maintenance cost.
E-creation	Energy cost for creating new chloride cells.
E-osmo	Total energy spent on osmoregulation.
motion-ratio	Fraction of metabolism allocated to swimming.
osmo-ratio	Fraction of metabolism allocated to osmoregulation.
M-max	Maximum size for scaling across the population.
max-age	Maximum age for scaling across the population.

4.4 Process Overview and Scheduling

1. Retrieve patch temperature ($^{\circ}\text{C}$) and verify safe environmental inputs.
2. Compute allometric, temperature, and age scalars (dimensionless).
3. Calculate metabolic rate for the current tick ((energy units) tick⁻¹).
4. Update metabolic accumulation (energy units).
5. Compute a reference baseline metabolic rate ((energy units) tick⁻¹).
6. Apply efficiency multipliers to swimming, digestion, osmoregulation, and lipid catabolism (%).

4.5 Design Concepts

Basic Principles: Metabolic rate combines allometric scaling, temperature responses, and age scaling consistent with metabolic ecology theory (Brown et al., 2004; West et al., 1997; Kleiber, 1932). The model uses a Q10 temperature function (?), Kleiber-type allometry (??), and age-dependent reduction in metabolic intensity (??).

Emergence: Energetic constraints emerge from interactions among size, temperature, and age. These constraints then shape movement, digestion, and osmoregulation.

Adaptation: Agents adapt by adjusting internal efficiencies that influence downstream behavioral and physiological submodels.

Objectives: Agents maintain adequate energy for survival and migration and balance energetic cost across processes.

Sensing: Agents sense the temperature of the patch they occupy.

Stochasticity: The function itself is deterministic but stochastic variation in individual size and age creates emergent metabolic variation.

Collectives: Scaling uses the maximum size and age of the entire population to normalize differences among individuals.

Observation: Outputs include metabolism-rate, total-metabolism, and efficiency multipliers.

4.6 Initialization

Variable	Initialized Value	Justification
Met-base	Computed using size, age, and temperature factors	Establishes a physiologically realistic baseline.
swim-base	Met-base times movement ratio	Allocates baseline metabolism for locomotion.
E-base	Met-base times 0.02	Maintenance cost of cellular metabolic activity.
E-creation	Met-base times 0.05	Cost of forming new chloride cells.
motion-ratio	0.7 normalized	Movement is energetically dominant.
osmo-ratio	0.3 normalized	Osmoregulation is energetically significant.
E-osmo	0	No osmoregulation cost at initialization.
metabolism-rate	Met-base	Baseline metabolism in the absence of environmental variation.
total-metabolism	0	Starts accumulation at zero.

4.7 Submodels

4.7.1 Initialize Metabolism

Baseline metabolic constant is computed at initialization for each agent using species-specific traits and fixed parameter values and remains constant across simulation runs:

Baseline metabolic constant is computed using:

$$Met_{base} = C_{base} \left(\frac{A_{mass}}{M_{max}} \right)^{0.75} \left[1 + S_{age} \left(1 - \frac{A_{age}}{Age_{max}} \right) \right] \left(\frac{T_{opt}}{T_{max}} \right) \quad (C.1)$$

Where:

- Met_{base} is the baseline metabolic rate used as the reference for all energetic scaling ((energy units) tick⁻¹).
- C_{base} is the constant baseline coefficient for metabolic initialization, set as 0.1 (dimensionless).
- A_{mass} is an agent's body mass used for allometric and energetic scaling (g).
- M_{max} is the maximum body mass across the population (g).
- S_{age} is the sensitivity parameter controlling age-related metabolic scaling, set to 0.5 (dimensionless).
- A_{age} is the agent's age in years (years).
- Age_{max} is the maximum age in the population (years).
- T_{opt} is the optimal temperature for metabolic performance (°C).
- T_{max} is the maximum temperature for agent survival, used for normalization (°C).

4.7.2 Energy Partitions

Baseline energy partitions are computed from the initialized metabolic base:

$$swim_{base} = Met_{base} \cdot Mt_{ratio} \quad (C.2)$$

Where:

- $swim_{base}$ is the baseline energetic cost of swimming ((energy units) tick⁻¹).
- Met_{base} is the baseline metabolic rate used as the reference for all energetic scaling ((energy units) tick⁻¹).

- Met_{ratio} is the ratio partitioning metabolic rate to baseline swimming cost (%).

$$E_{base} = Met_{base} \cdot 0.02 \quad (C.3)$$

Where:

- E_{base} is the baseline cell maintenance energy cost per tick ((energy units) tick⁻¹).
- Met_{base} is the baseline metabolic rate used as the reference for all energetic scaling ((energy units) tick⁻¹).

$$E_{creation} = Met_{base} \cdot 0.05 \quad (C.4)$$

Where:

- $E_{creation}$ is the energy cost of chloride cell creation scaled by body size ((energy units) tick⁻¹).
- Met_{base} is the baseline metabolic rate used as the reference for all energetic scaling ((energy units) tick⁻¹).

4.7.3 Metabolic Rate Calculation

At each tick, metabolism is recalculated using allometric, temperature, and age scaling. The 0.75 exponent follows quarter-power scaling relationships observed across taxa (???). This formulation reflects metabolic scaling relationships in teleost fishes and is applied consistently across species, with species-specific differences represented through parameter values (e.g., age, body mass, age structure, and thermal preferences).

Size Factor:

$$Size_f = \left(\frac{A_{mass}}{M_{max}} \right)^{0.75} \quad (C.5)$$

Where:

- $Size_f$ is the allometric scaling factor derived from relative body mass (dimensionless).
- A_{mass} is the agent's body mass used for allometric and energetic scaling (g).
- M_{max} is the maximum body mass across the population (g).

The Q10 formulation is widely used to represent the temperature sensitivity of metabolic rate in ectotherms (??).

The Temperature Factor is implemented as:

$$T_f = Q_{10}^{\left(\frac{T_{patch} - T_{opt}}{10}\right)} \quad (C.6)$$

Where:

- T_f is the temperature scaling factor applied to metabolism (dimensionless).
- Q_{10} is the temperature sensitivity coefficient in the Q10 formulation (dimensionless).
- T_{patch} is the local patch temperature experienced by the agent (°C).
- T_{opt} is the agent's optimal temperature for metabolic performance (°C).

Declines in mass-specific metabolic intensity with ontogeny and aging are incorporated using the following age-scaling factor (??):

The Age Factor is calculated as:

$$Age_f = 1 + \left(1 - \frac{A_{age}}{Age_{max}}\right) \cdot 0.5 \quad (C.7)$$

Where:

- $Age_{\{f\}}$ is the age-based scaling factor modifying metabolic intensity (dimensionless).
- A_{age} is the agent's age in years (years).
- Age_{max} is the maximum age in the population (years).

This multiplicative formulation follows the structure of metabolic ecology models in which metabolic rate scales as a function of body mass and temperature (?).

Total metabolic rate:

$$Met_{tick} = Met_{base} \cdot Size_f \cdot T_f \cdot Age_f \quad (C.8)$$

Where:

- Met_{tick} is the total metabolic rate for the current simulation tick ((energy units) tick⁻¹).
- Met_{base} is the baseline metabolic rate used as the reference for all energetic scaling ((energy units) tick⁻¹).

- $Size_f$ is the allometric scaling factor derived from relative body mass (dimensionless).
- T_f is the temperature scaling factor applied to metabolism (dimensionless).
- Age_f is the age-based scaling factor modifying metabolic intensity (dimensionless).

Metabolic accumulation:

$$Met_{total} = Met_{total} + Met_{tick} \quad (C.9)$$

Where:

- Met_{total} is the cumulative metabolic expenditure over time (energy units).
- Met_{tick} is the total metabolic rate for the current simulation tick ((energy units) tick⁻¹).

4.7.4 Efficiency Multipliers

A baseline metabolic reference is defined using:

- baseline size-factor = 1
- baseline age-factor = 1.5
- baseline temp-factor = 1

Thus:

$$M_{baseline} = Met_{base} \cdot 1 \cdot 1 \cdot 1.5$$

Efficiency is computed as:

$$Eff = \frac{Met_{tick}}{Met_{base}} \quad (C.10)$$

Where:

- Eff is the ratio of current to baseline metabolic rate (dimensionless).
- Met_{tick} is the total metabolic rate for the current simulation tick ((energy units) tick⁻¹).
- Met_{base} is the baseline metabolic rate used as the reference for all energetic scaling ((energy units) tick⁻¹).

This value is applied to:

- $swim_{Eff}$ is the efficiency multiplier applied to swimming energetics (dimensionless).
- $osmo_{Eff}$ is the efficiency multiplier applied to osmoregulatory costs (dimensionless).
- $lipid_{Eff}$ is the efficiency multiplier applied to lipid catabolism (%).

Efficiency dynamically scales the cost of movement, ion regulation, and lipid catabolism.

4.8 Netlogo Implementation

```
;; =====
;;   METABOLISM SUBMODEL
;; =====

globals [
  ;; global physiological constants
  Q10
]

patches-own [
  temperature
]

fish-own [
  ;; core state variables
  weight
  age-num
  optimal-temperature
  max-survival-temp

  ;; metabolic scalars
  Met-base
  metabolism-rate
  total-metabolism

  ;; efficiency multipliers
  swim-efficiency
  osmoregulation-efficiency
  lipid-catabolism-efficiency

  ;; energy partitioning
  swim-base
```

```

E-base
E-creation
E-osmo
motion-ratio
osmo-ratio

;; population normalization
M-max
max-age
]

;; =====
;;      INITIALIZE METABOLISM
;; =====

to initialize-metabolism
  let beta 0.75
  let age-sensitivity 0.5
  let base-constant 0.1

  ;; normalize population scaling
  set M-max max [weight] of fish
  set max-age max [age-num] of fish
  if M-max <= 0 [ set M-max 1 ]
  if max-age <= 0 [ set max-age 1 ]

  ;; scaling factors
  let size-factor (weight / M-max) ^ beta
  let age-factor 1 + age-sensitivity * (1 - (age-num / max-age))
  let temp-factor (optimal-temperature / max-survival-temp)

  ;; baseline metabolic constant
  set Met-base base-constant * size-factor * age-factor * temp-factor

  ;; energy partitions
  let motion-ratio-local 0.7
  let osmo-ratio-local 0.3
  let total-ratio (motion-ratio-local + osmo-ratio-local)

  set motion-ratio (motion-ratio-local / total-ratio)
  set osmo-ratio (osmo-ratio-local / total-ratio)

  set swim-base (Met-base * motion-ratio)
  set E-osmo 0

```

```

;; fixed fractions
set E-base      (Met-base * 0.02)
set E-creation (Met-base * 0.05)

;; initial metabolic state
set metabolism-rate Met-base
set total-metabolism 0
end

;; =====
;;          CALCULATE METABOLISM EACH TICK
;; =====

to calculate-metabolism
  let beta 0.75

  ;; repopulate scaling values
  set M-max max [weight] of fish
  set max-age max [age-num] of fish
  if M-max <= 0 [ set M-max 1 ]
  if max-age <= 0 [ set max-age 1 ]

  ;; get patch temperature safely
  let T-env [temperature] of patch-here
  if not is-number? T-env [ set T-env optimal-temperature ]

  ;; scaling factors
  let size-factor (weight / M-max) ^ beta
  let temp-factor Q10 ^ ((T-env - optimal-temperature) / 10)
  let age-factor 1 + (1 - (age-num / max-age)) * 0.5

  ;; metabolic rate (per tick)
  let M-total Met-base * size-factor * temp-factor * age-factor
  set metabolism-rate M-total
  set total-metabolism total-metabolism + M-total

  ;; baseline reference metabolism
  let baseline-size-factor 1
  let baseline-age-factor 1.5
  let baseline-temp-factor 1

  let baseline-metabolism Met-base *
    baseline-size-factor *
    baseline-temp-factor *

```

```
baseline-age-factor

;; efficiency multipliers
let efficiency M-total / baseline-metabolism

set swim-efficiency      efficiency
set osmoregulation-efficiency efficiency
set lipid-catabolism-efficiency efficiency
end
```


Chapter 5

Digestion

5.1 Overview

Digestion converts consumed prey mass into usable energy while transferring associated contaminants such as inorganic mercury and methylmercury from stomach contents into internal body burden pools. Within P-MEM, digestion is modeled as a metabolically mediated process, where digestive throughput is proportional to the agent's current metabolic rate and reflects established links between physiological state, energetic demand, and digestive processing in fishes, including gut evacuation and energy assimilation (??). As stomach contents are broken down, a fraction of biomass is converted into energy while biomass-associated contaminants are internalized, linking feeding, energetic gain, and contaminant uptake within a unified framework. When stomach contents are depleted, agents transition into foraging mode and begin drawing on lipid reserves.

5.2 Purpose

To simulate how migratory fish process ingested prey into energy and contaminants while linking digestive performance to metabolic rate, foraging behavior, and exposure risk. Digestion provides the energetic foundation required for movement, migration, and physiological processes while directly influencing contaminant accumulation.

5.3 Entities, State Variables, and Scales

5.3.1 Spatial and Temporal Scales

Spatial Unit: Patch (3 m x 3 m resolution)

Temporal Unit: 5 minute time steps (tick)

5.3.2 Global Variables

Global Variable	Definition
none required	Digestion uses internal agent state + metabolism-rate from metabolism submodel.

5.3.3 Patch Variables

Variable Name	Definition
None required	Digestion is an internal physiological process and does not depend on patch attributes.

5.3.4 Agent Variables

Variable Name	Definition
stomach-contents	Total mass of prey currently held in the stomach.
stomach-contents-Hg	Inorganic mercury contained within the stomach contents.
stomach-contents-MeHg	Methylmercury contained within the stomach contents.
digestion-rate	Rate of digestion equal to the current metabolic rate.
digestion-efficiency	Fraction of digested mass converted into usable energy.
energy	Total energetic reserves available to the agent.
gained-energy	Energy gained during the current tick from digestion.
mehg-foraging	Methylmercury released from stomach contents and absorbed during digestion.

Variable Name	Definition
hg-foraging	Inorganic mercury released from stomach contents during digestion.
mehg-foraging-total	Cumulative methylmercury risk through feeding.
hg-foraging-total	Cumulative inorganic mercury risk via digestion.
mehg-total	Total body burden of methylmercury risk.
hg-total	Total body burden of inorganic mercury risk.
foraging	Boolean indicating whether the agent is actively searching for prey.
lipid-loss	Boolean indicating when lipid reserves are used due to lack of stomach content.

5.4 Process Overview and Scheduling

1. Set digestion rate equal to the current metabolic rate ((energy units) tick⁻¹).
2. Calculate the mass digested during the current tick (g).
3. Compute the corresponding amounts of inorganic and methylmercury digested (ng).
4. Remove digested biomass and associated contaminants from stomach contents (g; ng).
5. Convert a fraction of digested biomass into usable energy ((energy units) tick⁻¹).
6. Add mercury and methylmercury to cumulative body burden pools (ng).
7. When stomach contents reach zero, set foraging to true and begin lipid use.

5.5 Design Concepts

Basic Principles: Digestion is modeled as a metabolic rate dependent process, where higher metabolic demand results in greater digestive throughput (?). Dietary MeHg uptake was represented as proportional to assimilated (digested) prey mass, consistent with mercury mass balance formulations that model uptake from food via an assimilation term (?).

Emergence: Patterns of energy gain and contaminant uptake emerge from interactions among foraging success, metabolism, digestion efficiency, and prey availability.

Adaptation: Agents respond to depleted stomach contents by transitioning into foraging mode and initiating lipid catabolism when necessary.

Objectives: Agents allocate available energy across movement, maintenance, and physiological processes to maximize reproductive success along the migration pathway while minimizing the risk of energetic depletion.

Sensing: Agents sense their own energetic state through stomach contents and energy levels.

Stochasticity: Stochasticity arises indirectly from variation in prey encounters or foraging success, although digestion itself is deterministic.

Collectives: Digestion occurs independently for each agent and does not require group level dynamics.

Observation: Outputs of interest include digested mass per tick, gained energy, contaminant uptake, and cumulative contaminant burdens.

5.6 Initialization

Variable	Initialized Value	Justification
stomach-contents	0	Agents begin without prey unless defined by scenario.
stomach-contents-Hg	0	No initial inorganic mercury from digestion.
stomach-contents-MeHg	0	No initial methylmercury from digestion.
digestion-efficiency	Species specific	Determines fraction of biomass converted to energy.
foraging	true	Agents begin in search mode when stomach is empty.

5.7 Submodels

5.7.1 Digestion Rate and Breakdown

Digestive throughput is driven by metabolic rate:

$$R_{digest} = Met_{tick} \quad (C.11)$$

Where:

- R_{digest} is the digestion rate of an individual agent (g tick⁻¹).
- Met_{base} is the baseline metabolic rate used as the reference for all energetic scaling ((energy units) tick⁻¹).

Digested mass per tick:

$$M_{digested} = M_{stomach} \cdot R_{digest} \cdot Eff_{digest} \quad (C.12)$$

Where:

- $M_{digested}$ is the biomass digested during the current tick (g).
- $M_{stomach}$ is the biomass currently stored in the stomach (g).
- R_{digest} is the digestion rate of an individual agent (g tick⁻¹).
- Eff_{digest} is the efficiency factor controlling digestion and filtration (%).

Proportional mercury assimilation:

$$MeHg_{digested} = \left(\frac{M_{digested}}{M_{stomach}} \right) \cdot MeHg_{stomach} \quad (C.13)$$

Where:

- $MeHg_{digested}$ is the methylmercury assimilated during digestion (ng).
- $M_{digested}$ is the biomass digested during the current tick (g).
- $M_{stomach}$ is the biomass currently stored in the stomach (g).
- $MeHg_{stomach}$ is the methylmercury currently stored in the stomach (ng).

Updated stomach contents:

$$M_{stomach} = M_{stomach} - M_{digested} \quad (C.14)$$

Where:

- $M_{digested}$ is the biomass digested during the current tick (g).
- $M_{stomach}$ is the biomass currently stored in the stomach (g).

5.7.2 Energy Conversion

Energy gained from digestion:

$$E_{gain} = M_{digested} \cdot 100 \quad (C.15)$$

Where:

- E_{gain} is the energy gained from digested biomass (energy units).
- $M_{digested}$ is the biomass digested during the current tick (g).

Updated energy pool:

$$E_{agent} = E_{agent} + E_{gain} \quad (C.16)$$

Where:

- E_{agent} is the total internal energy available to the agent (energy units).
- E_{gain} is the energy gained from digested biomass (energy units).

5.7.3 Contaminant Assimilation

Contaminants absorbed during digestion:

$$MeHg_{forage} = MeHg_{forage} + MeHg_{digested} \quad (C.17)$$

Where:

- $MeHg_{forage}$ is the cumulative methylmercury uptake via feeding (ng).
- $MeHg_{digested}$ is the methylmercury assimilated during digestion (ng).

Accumulated body burden:

$$MeHg_{total} = MeHg_{total} + MeHg_{digested} \quad (C.18)$$

Where:

- $MeHg_{total}$ is the cumulative methylmercury body burden (ng).
- $MeHg_{digested}$ is the methylmercury assimilated during digestion (ng).

5.7.4 Transition to Foraging

When stomach contents reach zero:

- lipid-loss is activated

This switches the prey to burn fat energy reserves instead of relying on stomach contents. This transition represents a shift from exogenous to endogenous energy use, consistent with anadromous fishes such as river herring, which reduce or cease feeding during spawning migration and rely on stored lipid reserves to support metabolic demands and movement (??).

5.8 Netlogo Implementation

```
;; =====
;; DIGESTION SUBMODEL
;; =====

fish-own [
  ;; digestion state
  stomach-contents
  stomach-contents-Hg
  stomach-contents-MeHg
  digestion-rate
  digestion-efficiency

  ;; contaminant tracking
  mehg-foraging
  hg-foraging
  mehg-foraging-total
  hg-foraging-total
  mehg-total
  hg-total

  ;; energy + biomass
  energy
  weight
  lipid-loss
  lipid-catabolism-efficiency
]

to digest
  ;; digestion tied to metabolic rate
  set digestion-rate metabolism-rate

  ;; nothing to digest
  if stomach-contents <= 0 [
    set foraging true
    set lipid-loss true
    stop
  ]
]
```

```

;; =====
;; DIGESTIVE BREAKDOWN
;; =====

;; amount of prey broken down this tick
let digested (stomach-contents * digestion-rate)

;; numerical guard to avoid divide-by-zero
if stomach-contents <= 0.000001 [ set digested 0 ]

let digested-Hg 0
let digested-MeHg 0

if digested > 0 and stomach-contents > 0 [
  set digested-Hg ((digested / stomach-contents) * stomach-contents-Hg)
  set digested-MeHg ((digested / stomach-contents) * stomach-contents-MeHg)
]

;; =====
;; UPDATE STOMACH CONTENTS
;; =====

set stomach-contents (max list 0 (stomach-contents - digested))
set stomach-contents-Hg (max list 0 (stomach-contents-Hg - digested-Hg))
set stomach-contents-MeHg (max list 0 (stomach-contents-MeHg - digested-MeHg))

;; =====
;; ENERGY CONVERSION
;; =====

let usable-energy (digested * digestion-efficiency * 40)
set energy energy + usable-energy

;; =====
;; STORE LIPID IF ENERGY SURPLUS
;; =====

if energy > 100 [
  let surplus (energy - 100)

  ;; convert surplus to lipid
  let lipid-added (surplus / 40)

  ;; apply storage efficiency
  set lipid-added (lipid-added * lipid-catabolism-efficiency)

```



```

;; update biomass
set weight weight + lipid-added

;; subtract energy equivalent
set energy energy - (lipid-added * 40)
]

;; =====
;; CONTAMINANT ASSIMILATION
;; =====

set meh-g-foraging      digested-MeHg
set hg-foraging         digested-Hg
set meh-g-foraging-total meh-g-foraging-total + digested-MeHg
set hg-foraging-total   hg-foraging-total + digested-Hg
set meh-g-total         meh-g-total + digested-MeHg
set hg-total            hg-total + digested-Hg

;; =====
;; SWITCH TO Lipid Loss IF EMPTY
;; =====

if stomach-contents <= 0 [
  set lipid-loss true
]
end

```


Chapter 6

Salinity Exposure

6.1 Overview

Osmoregulation allows migratory fish to maintain homeostasis by regulating internal ion concentrations in response to varying environmental salinities. Within the model, it is represented as a dynamic, state-dependent physiological process that links spatiotemporally explicit salinity exposure to ion-regulatory stress, chloride cell expression, and the metabolic energy costs of acclimation. Rather than assuming instantaneous equilibrium with surrounding salinity, agents experience osmotic stress based on local conditions and respond through time-dependent physiological adjustment (?).

6.2 Purpose

To simulate stress response to salinity changes for migratory fish in coastal systems by regulating chloride cell density and allocating energy toward ion-regulatory processes.

6.3 Entities, State Variables, and Scales

6.3.1 Spatial and Temporal Scales

- **Spatial Unit:** Patch (3 m x 3 m resolution)
- **Temporal Unit:** 5-minute time steps (`tick`)

6.3.2 Global Variables

	Global Variable	Definition
none required		Salinity exposure uses internal agent state

6.3.3 Patch Variables

Variable Name	Definition
Salinity S_{patch}	The salt concentration of a given patch, derived from hydrodynamic model inputs.

6.3.4 Agent Variables

Variable Name	Definition
acclimated-salinity S_{agent}	The salinity level the agent is currently acclimated to.
ionregulatory-stress I_{stress}	The level of stress an agent experiences when regulating ion balance due to osmotic difference.
chloride-density-min C_{min}	Minimum level of chloride cells, present even in low-stress conditions.
chloride-density-max C_{max}	Maximum level of chloride cells at high stress.
chloride-cell-density C	The current number of chloride cells expressed by the agent.
chloride-max-proliferation $R_{proliferation}$	The max number of chloride cells that can be expressed per time step.
chloride-cells-this-tick C_{tick}	The number of chloride cells created (or destroyed) in the current time step.
acclimation-rate α	The rate at which chloride cell density increases over time.
C-mid C_{mid}	The chloride cell density at which stress buffering is 50% effective.
time-since-last-osmoregulation t_{osmo}	The time elapsed since the last chloride cell regulation event.
Energy E_{agent}	The agent's total available energy for physiological functions.
E-osmo E_{osmo}	Total energy used for ion regulation (osmoregulation).
E-base E_{base}	The base energy cost per chloride cell.
E-creation $E_{creation}$	The energy cost for producing new chloride cells.
metabolic-max Met_{max}	Maximum metabolic cost for chloride cell creation.

6.4 Process Overview and Scheduling

1. Compute osmotic stress based on the difference between S_{patch} and S_{agent} , scaled from 1 to 10, where 1 represents the minimum salinity difference and 10 represents the maximum salinity difference observed across the simulation period.
2. Adjust chloride cell density depending on time since last osmoregulation (%).
3. Compute energy cost of osmoregulation (energy units).
4. Deduct energy expenditure from agent's energy pool (energy units).

6.5 Design Concepts

Basic Principles: The model is based on principles of physiological ecology and osmoregulatory energetics in teleost species (?). It draws from empirical findings on ion transport, chloride cell (ionocyte) proliferation, and the energetic costs of osmoregulation (e.g., ?; ?). This process is represented in the model as a delayed, rate-limited acclimation response, governed by osmotic stress, energy availability, and an acclimation interval (t_{osmo}), rather than being parameterized to a fixed empirical time-to-response.

Emergence: Ion-regulatory stress, chloride cell expression, and energy expenditure emerge from an agent's interaction with temporally and spatially variable salinity environments. These patterns are not pre-specified but arise dynamically through stress-driven, time-dependent physiological adjustment governed by acclimation dynamics and energy constraints.

Adaptation: Agents respond to osmotic stress by adjusting chloride cell density toward a stress-dependent target. The rate of adjustment is proportional to the difference between current and target chloride density and is constrained by a proliferation rate parameter, energy availability, and an acclimation interval (t_{osmo}), which limits how frequently updates occur. This process allows individuals to reduce internal-external salinity gradients and maintain ion homeostasis over time.

Objectives: Agents seek to support survival by reducing stress and avoiding excessive energy loss through regulating chloride cell expression.

Sensing: Agents sense local salinity (S_{patch}) and compare it with their acclimated salinity (S_{agent}). They also track their own energy state and time since last osmoregulation.

Stochasticity: Acclimation may vary with α , the acclimation rate parameter controlling how quickly an agent adjusts physiologically to salinity changes, which is assigned from a defined range for each individual to represent variability in osmoregulatory capacity across the population.

Observation: Outputs include S_{stress} (dimensionless, scaled 1–10), $C_{current}$ (dimensionless, relative chloride cell density rather than absolute cell count), E_{osmo} ((energy units) tick ¹), and E_{agent} (energy units). Energy is represented in relative energy units rather than absolute units (e.g., calories), allowing consistent scaling across processes without requiring direct empirical calibration. Chloride cell density is initialized at C_{min} and evolves dynamically toward a stress-dependent target over time.

6.6 Initialization

Variable	Initialized Value	Justification
S_{patch}	user-defined for data input	This input can be user-defined realistic data values or known spatial data.
S_{agent}	35 (psu)	Assumes agents start acclimated to marine environment.
I_{stress}	1	Acclimated agents have minimal stress levels.
C	50%	Starts with partial cell density, allowing for regulation depending on environmental conditions.
C_{min}	10%	A baseline level of chloride cells is necessary for basic osmoregulatory functions.
C_{max}	100%	Agents can't express more than 100% of cells.
α	0.0017 - 0.002	Osmolarity stabilization from Figure 3. in (Allen et al., 2009).
C_{mid}	50%	When cells are 50% density, stress buffering is 50% effective (Allen et al., 2009).
E_{agent}	100%	Agent starts with limited energy before migration.
E_{base}	Teleost (4%) Aptoerygian ()	Based on the branchial cost (Little et al., 2023; Kirschner, 1993).
Met_{max}	Teleost (3.5%)	Based on the intestinal and renal cost & size of agent (Little et al., 2023; Kirschner, 1993).
k	-0.75	Scaling component for body mass is negative (Kirschner, 1993) and follows Kleiber's Law.

6.7 Submodels

6.7.1 Salinity Stress

Salinity stress (S_{stress}) is calculated based on the difference between an agent's acclimated salinity and the ambient patch salinity, adjusted by the chloride cell buffering effect. This formulation represents a relative index of osmotic stress derived from salinity mismatch and buffering capacity, rather than a direct physiological measure (e.g., cortisol concentration):

$$S_{stress} = \frac{\log_{10}(1 + |S_{agent} - S_{patch}|) \cdot 10}{1 + e^{-2 \cdot (C_{current}/C_{mid})}} \quad (C.19)$$

Where:

- S_{stress} is the agent's physiological salinity stress index (dimensionless, scaled 1–10).
- S_{agent} is an agent's acclimated salinity (psu).
- S_{patch} is the environmental salinity at patch (psu).
- $C_{current}$ is the current chloride cell density (% , relative density).
- C_{mid} is the chloride density at 50% buffering effectiveness (% , relative density).

Stress is capped within the range $[1, 10]$, and may be reduced slightly over time if salinity remains stable and chloride density is sufficient:

$$S_{stress} = S_{stress} \cdot 0.98 \quad \text{if conditions are stable and } C > C_{min} \quad (C.20)$$

Where:

- S_{stress} is the agent's physiological salinity stress index (dimensionless, scaled 1–10).

Agents also slowly shift their acclimated salinity toward ambient salinity when conditions have been stable for several time steps (e.g., hours to days), consistent with documented time-dependent salinity acclimation in fishes (?).

$$S_{agent} = S_{agent} + (S_{patch} - S_{agent}) \cdot 0.02 \quad (C.21)$$

Where:

- S_{agent} is the agent's acclimated salinity (psu).
- S_{patch} is the environmental salinity at patch (psu).

6.7.2 Chloride Cell Density

Chloride cell proliferation is driven by the level of ion-regulatory stress the agent experiences when encountering a difference in salinity (?). In teleost fishes, ion transport capacity and chloride cell function are regulated through changes in transporter activity and cell abundance in response to environmental salinity and endocrine signaling, rather than instantaneous adjustment. Accordingly, the greater the stress, the higher the target chloride density the agent attempts to reach, up to a maximum threshold. Chloride cell density is represented as a relative value expressed as a percentage of maximum capacity, where 10%

corresponds to the minimum density (stress level 1) and 100% corresponds to the maximum density (stress level 10). Agents adjust their chloride cell density gradually over time based on their current ion-regulatory stress and acclimation status, with the rate of adjustment constrained by proliferation dynamics, energy availability, and the acclimation interval (t_{osmo}). Chloride cells are not adjusted unless the agent's energy exceeds 25%.

The chloride cell density is based on stress:

$$C_{target} = C_{min} + (C_{max} - C_{min}) \cdot \left(\frac{S_{stress}}{10} \right) \quad (C.22)$$

Where:

- C_{target} is the target chloride density based on stress (% , relative density).
- C_{min} is the minimum chloride cell density (% , relative density).
- C_{max} is the maximum chloride cell density (% , relative density).
- S_{stress} is the agent's physiological salinity stress index (dimensionless, scaled 1–10).

If salinity conditions have remained stable for an extended period (e.g., 288 ticks), (C_{target}) is slightly reduced to reflect partial down regulation of chloride cells due to long-term acclimation:

$$C_{target} = C_{target} \cdot 0.99 \quad \text{if stable} \quad (C.23)$$

Where:

- C_{target} is the target chloride density based on stress (% , relative density).

Chloride cell density then approaches the target using a double-rate adjustment and capped maximum rate of change:

$$\Delta C = (C_{target} - C_{current}) \cdot (2 \cdot R_{proliferation}) \quad (C.24)$$

Where:

- ΔC is the updated chloride density after regulation (% , relative density).
- C_{target} is the target chloride density based on stress (% , relative density).
- $C_{current}$ is the current chloride cell density (% , relative density).
- $R_{proliferation}$ is the chloride cell proliferation rate ((%) tick⁻¹, relative density per tick).

$$\Delta C = \max(-R_{max}, \min(R_{max}, \Delta C)) \quad (C.25)$$

Where:

- ΔC is the updated chloride density after regulation (% , relative density).

- R_{max} is the maximum allowable chloride change per update ((%) tick¹, relative density per tick).

If the agent has low energy ($\leq 50\%$), the adjustment rate is halved:

$$\Delta C = \Delta C \cdot 0.5 \quad \text{if energy is low} \quad (\text{C.26})$$

Where:

- ΔC is the updated chloride density after regulation (% , relative density).

Finally, the chloride cell density is updated and constrained between C_{min} and C_{max} :

$$C_{new} = \max(C_{min}, \min(C_{max}, C_{current} + \Delta C)) \quad (\text{C.27})$$

Where:

- C_{new} is the updated chloride density after regulation (% , relative density).
- C_{min} is the minimum chloride cell density (% , relative density).
- $C_{current}$ is the current chloride cell density (% , relative density).
- C_{max} is the maximum chloride cell density (% , relative density).
- ΔC is the updated chloride density after regulation (% , relative density).

This ensures that the agent does not overshoot the physiologically realistic limit of chloride cell density, while still responding to salinity stress.

Chloride density is only recalculated after a given acclimation interval:

$$t_{osmo} \geq \alpha^{-1} \quad (\text{C.28})$$

Where:

- t_{osmo} is the time since last osmoregulation update (ticks).
- α is the acclimation rate constant (dimensionless/tick).

After updating, the acclimation timer is reset:

$$t_{osmo} = 0 \quad (\text{C.29})$$

Where:

- t_{osmo} is the time since last osmoregulation update (ticks).

This prevents agents from recalculating chloride density every time step and allows for controlled, realistic responses to prolonged stress and salinity changes.

Osmoregulation Energy

This formulation follows established allometric scaling relationships in ecological physiology, where metabolic and physiological processes scale with body mass (g) according to a power-law relationship (e.g., ?; ?; ?; ?).

Metabolic cost related to mass (g):

$$E_{creation} = Met_{max} * (\frac{A_{mass}}{M_{max}})^k \quad (C.30)$$

Where:

- $E_{creation}$ is the energy cost of chloride cell creation scaled by body size ((energy units) tick⁻¹).
- Met_{max} is the maximum metabolic cost parameter for osmoregulation ((energy units) tick⁻¹).
- A_{mass} is the agent's body mass used for allometric and energetic scaling (g).
- M_{max} is the maximum body mass across the population (g).
- k is the energy scaling exponent (dimensionless).

Energy required for ion regulation:

$$E_{osmo} = (E_{base} \cdot C) + (E_{creation} \cdot C_{tick}) \quad (C.31)$$

Where:

- E_{base} is the baseline cell maintenance energy cost per tick ((energy units) tick⁻¹).
- $C_{current}$ is the current chloride cell density (% relative density).
- $E_{creation}$ is the energy cost of chloride cell creation scaled by body size ((energy units) tick⁻¹).
- C_{tick} is the relative percent of new chloride cells produced in a tick (% relative density).

6.7.3 Energy Balance

Agents balance energy to osmoregulate with total energy allowance:

$$E_{agent} = E_{agent} - E_{osmo} \quad (C.32)$$

Where:

- E_{agent} is the total internal energy available to the agent (energy units).
- E_{osmo} is the energy cost of osmoregulation (energy units).

6.8 Netlogo Implementation

```
;; =====
;; SALINITY EXPOSURE AND OSMOREGULATION SUBMODEL
;; =====

globals [
  stable-time          ;; tracks duration of stable salinity
  salinity-threshold   ;; threshold difference to count as "stable"
]

patches-own [
  salinity             ;; environmental salinity (psu)
]

fish-own [
  ;; --- Salinity Physiology ---
  acclimated-salinity
  ionregulatory-stress  ;; 1-10 scale

  ;; --- Chloride Cell Physiology ---
  chloride-density-min
  chloride-density-max
  chloride-cell-density
  chloride-max-proliferation
  chloride-cells-this-tick
  C-mid

  ;; --- Acclimation Timing ---
  acclimation-rate
  time-since-last-osmoregulation

  ;; --- Energetics ---
  energy
  E-osmo
  E-base
  E-creation
  Met-max

  ;; --- Size Scaling ---
  weight
]
```

```

]

;; =====
;; MAIN OSMOREGULATION PROCEDURE
;; =====

to osmoregulate
  calculate-ionregulatory-stress
  regulate-chloride-cell-density
  calculate-osmoregulation-energy
end

;; =====
;; STEP 1 - CALCULATE ION-REGULATORY STRESS
;; =====

to calculate-ionregulatory-stress
  let prev-sal [salinity] of previous-patch
  let patch-sal [salinity] of patch-here
  let agent-sal acclimated-salinity

  ;; --- Base salinity difference ---
  let sal-diff abs (agent-sal - patch-sal)

  ;; --- Core salinity-based stress function ---
  let salinity-stress salinity-to-stress sal-diff chloride-cell-density ionregulatory-s

  ;; --- Salinity stability, used for acclimation ---
  let sal-change abs (patch-sal - prev-sal)

  if sal-change < salinity-threshold [
    set stable-time stable-time + 1
  ]
  if sal-change >= salinity-threshold [
    set stable-time 0
  ]

  ;; --- Slow acclimation when salinity is stable ---
  if stable-time > 4 and chloride-cell-density > chloride-density-min [
    set salinity-stress salinity-stress * 0.98
    set acclimated-salinity acclimated-salinity +
      ((patch-sal - acclimated-salinity) * 0.02)
  ]

```

```

;; --- Chloride buffering ---
let buffer-reduct calculate-stress-reduction-chloride-function chloride-cell-density

;; Buffering only applies above minimum chloride levels
if chloride-cell-density > chloride-density-min [
  set ionregulatory-stress max (list 1 (salinity-stress - buffer-reduct * 0.8))
]

if chloride-cell-density <= chloride-density-min [
  set ionregulatory-stress salinity-stress
]
end

;; =====
;; STEP 2 - REGULATE CHLORIDE CELL DENSITY
;; =====

to regulate-chloride-cell-density
  ;; energetic constraint
  if energy <= 25 [ stop ]

  set time-since-last-osmoregulation time-since-last-osmoregulation + 1

  ;; wait until acclimation interval
  if time-since-last-osmoregulation < acclimation-rate [ stop ]

  ;; --- Target chloride cell density ---
  let target-dens chloride-density-min +
    ((chloride-density-max - chloride-density-min) *
     (ionregulatory-stress / 10))

  ;; down-regulate if long-term stability
  if stable-time > 288 [
    set target-dens (target-dens * 0.99)
  ]

  ;; constrain target to bounds
  set target-dens max (list chloride-density-min
                        min (list chloride-density-max target-dens))

  ;; --- Adjustment rate ---
  let change (target-dens - chloride-cell-density) *
    (2 * chloride-max-proliferation)

```

```

let max-change ((chloride-density-max - chloride-density-min) *
                 chloride-max-proliferation)

;; clamp change
set change max (list (-1 * max-change) (min (list max-change change)))

;; energy-dependent reduction
if energy <= 50 [
  set change (change * 0.5)
]

;; apply change
set chloride-cell-density chloride-cell-density + change
set chloride-cell-density max (list chloride-density-min
                                   min (list chloride-density-max chloride-cell-density))

set chloride-cells-this-tick abs change
set time-since-last-osmoregulation 0
end

;; =====
;; STEP 3 - OSMOREGULATION ENERGY COST
;; =====

to calculate-osmoregulation-energy
  let maintenance (E-base * chloride-cell-density)
  let creation     (E-creation * chloride-cells-this-tick)

  set E-osmo (maintenance + creation)
  set energy max (list 0 (energy - E-osmo))
end

;; =====
;; CHLORIDE BUFFERING FUNCTION
;; =====

to-report calculate-stress-reduction-chloride-function [C]
  ;; logistic-like buffering curve
  let reduction 5 * (1 - exp(C * -0.5 / C-mid))
  report reduction
end

```

```

;; =====
;; SALINITY → STRESS FUNCTION
;; =====

to-report salinity-to-stress [diff C prev-stress]
  let scaled (log (1 + abs diff) 10) * 10
  let buffer (1 + exp(-2 * (C / C-mid)))
  let result (scaled / buffer)

  ;; constrain to 1-10
  set result max (list 1 (min (list 10 result)))

  ;; acclimation effect
  if stable-time > 288 [
    set result (result * 0.99)
  ]

  report result
end

;; =====
;; SIZE-SCALED CREATION COST
;; =====

to calculate-E-creation
  let M-max max [weight] of fish
  if M-max <= 0 [ set M-max weight ]

  let beta -0.75
  set E-creation Met-max * ((weight / M-max) ^ beta)
end

```


Chapter 7

Migration Cues

7.1 Overview

This submodel simulates how migratory fish detect environmental cues that signal the onset of migration, based on the interaction between photoperiod and temperature, which are widely recognized as primary drivers of migration timing in anadromous fishes. Photoperiod provides a seasonal signal regulating physiological readiness, while temperature modifies the timing of migration initiation and can act as a releasing factor (e.g., ?; ?). In this model, migration begins when individuals experience a coincident decline in both daylength and temperature ($\Delta L < 0$ and $\Delta T < 0$ on the same day), representing a simplified autumnal cue window. These events are classified as cue days, during which individuals stochastically accumulate the probability of initiating migration. For predators such as striped bass, migration probability is additionally linked to alewife presence, integrating species interactions into migration onset.

Rather than assuming a fixed threshold, this formulation represents migration cues as periods of coincident decline in environmental conditions, capturing seasonal transitions that trigger movement. This cue formulation is applied specifically to upstream migration and is site-specific, requiring evaluation and potential recalibration for other systems. Downstream migration is not explicitly cued and instead emerges from the model following completion of migration objectives such as spawning or foraging.

7.2 Purpose

The purpose of this submodel is to mechanistically trigger migration using **real, observed environmental phenology** rather than arbitrary or date-based rules. This framework captures how:

- Changes in Δ Temperature and Δ Daylength structure seasonality
- Migration timing emerges pragmatically rather than being fixed
- Fish respond to external conditions and species interactions
- Environmental preprocessing in R feeds into NetLogo behavioral rules

This allows migration in the model to originate from measurable ecological drivers instead of hard-coded thresholds.

7.3 Entities, State Variables, and Scales

7.3.1 Spatial and Temporal Scales

- **Spatial Unit:** Patch (3 m x 3 m resolution)
- **Temporal Unit:** 5-minute time steps (`tick`)

7.3.2 Global Variables

Global Variable	Definition
cue-table	Lookup table containing Δ Temperature, Δ Daylength, their smoothed values, and Boolean coincident-decline flags (from R preprocessing)
cue-days	List of day-of-year values where $\Delta T < 0$ and $\Delta L < 0$ simultaneously (identified in R)
ΔT	Daily rate of change in temperature imported from R (<code>dT_per_day_f</code> or smoothed <code>dT_smooth</code>)
ΔL	Daily rate of change in photoperiod imported from R (<code>dL_per_day_min</code> or smoothed <code>dL_smooth</code>)
cue-active?	Whether the current day-of-year corresponds to a coincident decline day
migration-trigger?	Whether environmental signals are strong enough to begin building migration probability

7.3.3 Patch Variables

Migration cues do not depend on patch-level conditions. No patch-scale variables are required.

7.3.4 Agent Variables

Variable	Definition
migration-trigger?	Whether the agent has entered a window where migration probability begins to accumulate
migration-probability	The cumulative probability of initiating migration, updated each tick once the trigger is active
start-migration?	Whether the individual has begun active migration

7.4 Process Overview and Scheduling

1. Environmental Inputs (from R)

External R preprocessing computes:

- Δ Temperature per day
- Δ Daylength per day
- Smoothed 3-day rolling means
- Boolean `both_negative` indicating coincident decline
- Exports full dataset as "`cue_dataset_full.csv`"

This dataset defines the environmental basis for triggering migration.

2. When a cue day is detected, activate the migration cue, defined by coincident declines in the rate of change of temperature and photoperiod, and enable migration readiness.
3. For agents with an active migration trigger, individual migration probability is incremented at each simulation time step (tick).
4. When an agent's migration probability exceeds a stochastic threshold, initiate migration.

7.5 Design Concepts

Basic Principles: The mechanism reflects literature on the interaction between photoperiod and temperature as drivers of migration timing in anadromous fishes, where migration responses emerge from the combined influence of seasonal and short-term environmental signals (e.g., ?; ?; ?).

Emergence: Migration timing varies among individuals (?) and is represented in the model through stochastic variation in the accumulation of migration probability, resulting in asynchronous migration initiation across agents.

Adaptation: Agents transition from staging to active migration in response to environmental cue activation and internal readiness state.

Stochasticity: Stochasticity enters through probabilistic accumulation of migration readiness and random threshold exceedance for migration initiation.

Observation: Key outputs include cue activation timing, individual migration probability trajectories, and the proportion of agents initiating migration through time.

7.6 Initialization

Variable	Initial Value	Justification
cue-days	Extracted from CSV	Based on empirical environmental data
migration-probability	0	Ensures probability is accumulated only within cue window
migration-trigger?	false	No cue until ΔT and ΔL decline
cue-active?	false	Depends on DOY
start-migration?	false	Fish begin in staging mode

7.7 Submodels

7.7.1 Identifying Cue Days (Preprocessed in R)

$$\Delta T < 0, \Delta L < 0 \quad (\text{C.33})$$

Where:

- ΔT is the change in temperature used to flag cue days.
- ΔL is the change in photoperiod proxy used to flag cue days.

The preprocessing script:

1. Computes ΔT and ΔL
2. Applies smoothing
3. Flags coincident declines

7.7.2 Activation of Migration Cue

At each simulation day, the model checks whether the current day-of-year corresponds to a predefined cue day. If true, the migration cue is activated:

$$cue-active? = \text{true} \quad (\text{C.34})$$

Where:

- *cue-active?* is the indicator that migration cue is active.

This represents the environmental system state in which the cue has occurred, after which agents respond by transitioning to an internal readiness state:

$$migration-trigger? = \text{true} \quad (\text{C.35})$$

Where:

- *migration-trigger?* is the indicator of migration readiness.

This marks the onset of migration preparedness for all agents, where an individual has sensed the cue and transitions to an internally primed state. In migratory organisms, environmental signals are sensed first and then establish readiness, after which movement is initiated only when conditions are favorable rather than immediately upon cue detection (?).

7.7.3 Accumulation of Migration Probability

For each agent with an active migration trigger, migration probability increases stochastically each tick:

$$P_{t+1} = P_t + (0.00001 + U(0, 0.00004)) \quad (\text{C.36})$$

Where:

- P_t is the individual migration probability at time t .

Migration probability is capped at:

$$P_t \leq 1 \quad (\text{C.37})$$

Where:

- P_t is the individual migration probability at time t .

This formulation produces gradual increase into migration and individual variation in migration timing.

7.7.4 Initiate Migration

An agent initiates migration when its migration probability exceeds a stochastic threshold:

$$\text{random-float}(1) < P_t \quad (\text{C.38})$$

Where:

- P_t is the individual migration probability at time t .

When this condition is met:

$$\text{start-migration?} = \text{true} \quad (\text{C.39})$$

Where:

- *start-migration?* is the indicator that migration has begun.

The agent transitions from staging behavior to directed migratory movement.

7.8 Netlogo Implementation

```
;; -----
;; GLOBALS
;; -----
globals [
  cue-table
  cue-days

  migration-trigger?
  cue-active?
]

;; -----
;; TURTLE VARIABLES
;; -----
turtles-own [
  migration-probability
  start-migration?
]

;; -----
```

```

;; LOAD CUE DATA (CSV FROM R)
;; -----
to load-cue-csv
  set cue-table table:make
  let raw csv:from-file "inputs/cue_dataset_full.csv"

  let header first raw
  let rows but-first raw

  foreach rows [
    r ->
      let doy      item 7 r
      let dTsm     item 21 r
      let dLsm     item 22 r
      let negflag  item 24 r

      table:put cue-table doy (list dTsm dLsm negflag)
  ]
end

;; -----
;; IDENTIFY CUE DAYS ( $\Delta T < 0$  AND  $\Delta L < 0$ )
;; -----
to find-migration-days
  set cue-days []

  foreach sort table:keys cue-table [
    d ->
      let vals table:get cue-table d
      let negflag item 2 vals

      if (negflag = "TRUE" or negflag = true) [
        set cue-days lput d cue-days
      ]
  ]
end

;; -----
;; UPDATE MIGRATION CUE (DAILY)
;; -----
to update-migration-cue
  ifelse member? day cue-days [
    set cue-active? true
    set migration-trigger? true
  ] [

```

```
    set cue-active? false
  ]
end

;; -----
;; ALEWIFE MIGRATION PROBABILITY ACCUMULATION
;; -----
to calc-migration-probability
  if migration-trigger? [
    set migration-probability
      migration-probability
      + (0.00001 + random-float 0.00004)

    if migration-probability > 1 [
      set migration-probability 1
    ]
  ]
end

to migrate
  if random-float 1 <= migration-probability
  set start-migration? true
end
```


Chapter 8

Contaminant Exposure

8.1 Overview

The contaminant exposure submodel simulates mercury and methylmercury uptake across the gills for migratory fish moving through spatially and temporally variable contamination fields. Exposure risk is governed by contaminant concentration, suspended particulate matter (SPM), and metabolic rate, with concentrations normalized across the spatial domain and depth-averaged SPM modulating risk to reflect particle-bound transport. Agents accumulate both instantaneous and cumulative exposure, allowing estimation of bioaccumulation risk over the migration season. This formulation reflects established mechanisms of methylmercury (MeHg) uptake in fishes, where exposure during active movement occurs by branchial uptake across the gills (??). The gill epithelium serves as the primary interface for this exchange, with uptake regulated by diffusion, ion transport, and local water chemistry, and increasing with metabolic demand as elevated respiration enhances water flux across the gills; physiological stress can further modify uptake through changes in ion regulation and membrane permeability (?).

8.2 Purpose

To evaluate mercury and methylmercury exposure and bioaccumulation risk for migratory fishes by integrating patch level contamination, ionregulatory stress, depth averaged suspended particulate matter, and metabolism driven uptake processes in estuarine and coastal systems.

8.3 Entities, State Variables, and Scales

8.3.1 Spatial and Temporal Scales

Spatial Unit: Patch (3 m x 3 m resolution)

Temporal Unit: 5 minute time steps (tick)

8.3.2 Global Variables

Variable	Initialized Value	Justification
MeHg-threshold	15 ug per kg	Screening threshold for methylmercury (??)
Hg-threshold	150 ug per kg	NOAA guideline for mercury contamination (??)
min-Hg, max-Hg	Calculated from input layers	Required for Hg normalization across domain
min-MeHg, max-MeHg	Calculated from input layers	Required for MeHg normalization across domain

8.3.3 Patch Variables

Variable Name	Definition
mercury	Mercury concentration at this patch.
methylmercury	Methylmercury concentration at this patch.
SPM	Depth averaged suspended particulate matter at this patch.
Hg-exp-risk-alewife	Patch level mercury exposure risk from alewives.
Hg-exp-risk-stripedbass	Patch level mercury exposure risk from striped bass.
MeHg-exp-risk-alewife	Patch level methylmercury risk from alewives.
MeHg-exp-risk-stripedbass	Patch level methylmercury risk from striped bass.

8.3.4 Agent Variables

Variable Name	Definition
metabolism-rate	Current metabolic rate used to scale uptake risk.
hg-exposure-duration	Number of ticks spent above the mercury threshold.
mehg-exposure-duration	Number of ticks spent above the MeHg threshold.
hg-uptake-risk	Instantaneous mercury uptake risk per tick.
mehg-uptake-risk	Instantaneous methylmercury uptake risk per tick.
hg-total	Cumulative mercury uptake.
mehg-total	Cumulative methylmercury uptake.
hg-exposure-total	Total non normalized Hg exposure.
mehg-exposure-total	Total non normalized MeHg exposure.
hg-exposure-total-normalized	Sum of normalized Hg exposure values.
mehg-exposure-total-normalized	Sum of normalized MeHg exposure values.

8.4 Process Overview and Scheduling

1. Retrieve methylmercury (ng g^{-1}), and depth-averaged SPM values from the current patch ($\frac{\text{kg}}{\text{m}^3}$).
2. Normalize patch contaminants between 0 and 1.
3. Compare patch levels to stress thresholds.
4. Update exposure durations (ticks).
5. Compute uptake risk using metabolism ((energy units) tick^{-1}), ion-regulatory stress (dimensionless), patch methylmercury (ng g^{-1}), and SPM ($\frac{\text{kg}}{\text{m}^3}$).
6. Add uptake risk to cumulative body burden (dimensionless), initialized at zero but configurable based on user-defined initial conditions.
7. Update patch-level species-specific risk metrics.

8.5 Design Concepts

Basic Principles: Contaminant uptake follows toxico-kinetic principles, where absorption is driven by dissolved methylmercury concentrations and mediated by physiological processes, including metabolic rate and stress, that regulate uptake, retention, and elimination (e.g., ?; ?). SPM represents the hydrodynamic transport medium for sediment-bound contaminants and regulates the availability of dissolved MeHg in the water column, modulating exposure by influencing the concentration of dissolved contaminants available for branchial uptake.

Emergence: Exposure and risk patterns emerge from individual movement across heterogeneous contamination fields, coupled with metabolic demand (energy units per tick) and stress-driven physiology (dimensionless).

Adaptation: Agents do not actively avoid contamination, but their internal physiology modulates uptake in response to metabolic changes.

Objectives: This model does not give agents risk minimization behavior. Instead, risk is assessed to evaluate plausible exposure pathways and cumulative toxic exposure.

Sensing: Agents track patch-level mercury, methylmercury, and SPM, along with internal metabolic state, to quantify contaminant exposure and flux; this is a model construct for accounting purposes and does not represent biological sensing.

Stochasticity: Variation in spatial contaminant fields or individual characteristics and state creates stochastic variation in exposure.

Observation: Outputs include instantaneous exposure risk, cumulative exposure risk, normalized exposure totals, and patch level species specific contributions.

8.6 Initialization

Variable	Initialized Value	Justification
mercury	Input layer	Hydrodynamic or field derived contaminant layer.
methylmercury	Input layer	Same as above.
SPM	Depth averaged input	Reflects hydrodynamic particle transport.

8.7 Submodels: Mercury Exposure and Uptake

8.7.1 Normalization

$$MeHg_{normalized} = \frac{MeHg_{patch} - MeHg_{min}}{MeHg_{max} - MeHg_{min}} \quad (C.40)$$

Where:

- $MeHg_{normalized}$ is the normalized methylmercury concentration (dimensionless).
- $MeHg_{patch}$ is the patch methylmercury concentration at the agent location (ng g^{-1}).
- $MeHg_{min}$ is the minimum methylmercury value for normalization, relative to minimum contaminant concentration observes in system (ng g^{-1}).
- $MeHg_{max}$ is the maximum methylmercury value for normalization, relative to maximum contaminant concentration observes in system (ng g^{-1}).

Clamped to the range 0 to 1.

8.7.2 Exposure Duration

$$MeHg_{exp} = MeHg_{patch} + 1 \quad \text{if } MeHg_{patch} > MeHg_{threshold} \quad (C.41)$$

Where:

- $MeHg_{exp}$ is the Duration of exposure above threshold (ticks).
- $MeHg_{patch}$ is the patch-level methylmercury used in feeding uptake (ng g^{-1}).
- $MeHg_{threshold}$ is the threshold concentration for exposure duration tracking, initialized at the Effects Range-Low for MeHg (ERL = 15 ng g^{-1}) (?).

8.7.3 Uptake Risk

SPM acts as a scaling factor on exposure risk, representing increased availability of dissolved MeHg associated with elevated SPM concentrations.

$$MeHg_{risk} = Met_{tick} \cdot MeHg_{normalized} \cdot (1 + S_{stress}) \cdot (1 + SPM) \quad (C.42)$$

Where:

- $MeHg_{risk}$ is an agent's instantaneous methylmercury exposure risk (dimensionless).
- Met_{tick} is the total metabolic rate for the current simulation tick (energy units).

- $MeHg_{normalized}$ is the normalized methylmercury concentration (dimensionless).
- S_{stress} is an agent's physiological salinity stress index (dimensionless).
- SPM is the suspended particulate matter (SPM) concentration on the patch ($\frac{kg}{m^3}$).

8.7.4 Cumulative Uptake

$$MeHg_{total} = MeHg_{total} + MeHg_{risk} \quad (C.43)$$

Where:

- $MeHg_{total}$ is an agent's cumulative methylmercury body burden (dimensionless).
- $MeHg_{risk}$ is an agent's instantaneous methylmercury exposure risk (dimensionless).

Patch-level and individual-level storage based on species for cumulative duration risk experienced:

$$MeHg_{exp-risk} = MeHg_{exp-risk} + MeHg_{risk} \quad (C.44)$$

$$MeHg_{exp-patch} = MeHg_{exp-patch} + MeHg_{risk} \quad (C.45)$$

Where:

- $MeHg_{exp-risk}$ is the cumulative exposure-weighted risk per individual (dimensionless).
- $MeHg_{exp-patch}$ is the patch-level cumulative exposure contribution (dimensionless).

8.8 Netlogo Implementation

```
globals [
  ;; Thresholds
  MeHg-threshold
  Hg-threshold

  ;; Domain-wide normalization bounds
  min-Hg
  max-Hg
  min-MeHg
  max-MeHg
]
```

```
patches-own [  
  ;; Contaminant inputs  
  mercury  
  methylmercury  
  SPM  
  
  ;; Species-specific instantaneous risk  
  Hg-exp-risk-alewife  
  Hg-exp-risk-stripedbass  
  MeHg-exp-risk-alewife  
  MeHg-exp-risk-stripedbass  
  
  ;; Species-specific cumulative patch totals  
  Hg-exp-risk-total-alewife  
  Hg-exp-risk-total-stripedbass  
  MeHg-exp-risk-total-alewife  
  MeHg-exp-risk-total-stripedbass  
]  
  
turtles-own [  
  ;; Physiology  
  metabolism-rate  
  ionregulatory-stress  
  
  ;; Exposure durations  
  hg-exposure-duration  
  mehg-exposure-duration  
  
  ;; Instantaneous uptake  
  hg-uptake-risk  
  mehg-uptake-risk  
  
  ;; Cumulative uptake  
  hg-total  
  mehg-total  
  
  ;; Cumulative exposure raw  
  hg-exposure-total  
  mehg-exposure-total  
  
  ;; Cumulative normalized exposure  
  hg-exposure-total-normalized  
  mehg-exposure-total-normalized  
]
```

```

;; =====
;; MERCURY EXPOSURE
;; =====

to mercury-contamination

  ;; -----
  ;; Retrieve patch contaminant and SPM values
  ;; -----
  let Hg-current [mercury] of patch-here
  let SPM-current [SPM] of patch-here

  if not is-number? Hg-current [ set Hg-current 0 ]
  if not is-number? SPM-current [ set SPM-current 0 ]

  ;; -----
  ;; Normalize Hg concentration (0-1)
  ;; -----
  let Hg-normalized 0
  if (max-Hg - min-Hg) > 0 [
    set Hg-normalized (Hg-current - min-Hg) / (max-Hg - min-Hg)
  ]

  ;; clamp
  set Hg-normalized max list 0 (min list 1 Hg-normalized)

  ;; -----
  ;; Exposure duration: threshold exceedance
  ;; -----
  if Hg-current > Hg-threshold [
    set hg-exposure-duration hg-exposure-duration + 1
  ]

  ;; -----
  ;; Uptake risk (toxicokinetic rule)
  ;; -----
  let stress-factor (1 + ionregulatory-stress)
  let spm-factor    (1 + SPM-current)

  set hg-uptake-risk (metabolism-rate * Hg-normalized * stress-factor * spm-factor)

  ;; no negative risk ever
  if hg-uptake-risk < 0 [ set hg-uptake-risk 0 ]

  ;; -----

```



```

;; Add to cumulative body burden
;; -----
set hg-total hg-total + hg-uptake-risk

;; -----
;; Track cumulative raw & normalized exposure
;; -----
set hg-exposure-total          hg-exposure-total + Hg-current
set hg-exposure-total-normalized hg-exposure-total-normalized + Hg-normalized

;; -----
;; Patch-level risk recording (species specific)
;; -----
let patch-risk hg-uptake-risk

if breed = alewives [
  ask patch-here [
    set Hg-exp-risk-alewife      patch-risk
    set Hg-exp-risk-total-alewife (Hg-exp-risk-total-alewife + patch-risk)
  ]
]

if breed = stripedbass [
  ask patch-here [
    set Hg-exp-risk-stripedbass      patch-risk
    set Hg-exp-risk-total-stripedbass (Hg-exp-risk-total-stripedbass + patch-risk)
  ]
]

end

;; =====
;; METHYLMERCURY EXPOSURE
;; =====

to methylmercury-contamination

;; -----
;; Retrieve patch contaminant and SPM values
;; -----
let MeHg-current [methylmercury] of patch-here
let SPM-current  [SPM] of patch-here

if not is-number? MeHg-current [ set MeHg-current 0 ]

```

```

if not is-number? SPM-current [ set SPM-current 0 ]

;; -----
;; Normalize MeHg concentration (0-1)
;; -----
let MeHg-normalized 0
if (max-MeHg - min-MeHg) > 0 [
  set MeHg-normalized (MeHg-current - min-MeHg) / (max-MeHg - min-MeHg)
]

;; clamp
set MeHg-normalized max list 0 (min list 1 MeHg-normalized)

;; -----
;; Exposure duration: threshold exceedance
;; -----
if MeHg-current > MeHg-threshold [
  set mehg-exposure-duration mehg-exposure-duration + 1
]

;; -----
;; Uptake risk
;; -----
let stress-factor (1 + ionregulatory-stress)
let spm-factor (1 + SPM-current)

set mehg-uptake-risk (metabolism-rate * MeHg-normalized * stress-factor * spm-factor)

if mehg-uptake-risk < 0 [ set mehg-uptake-risk 0 ]

;; -----
;; Add to cumulative body burden
;; -----
set mehg-total mehg-total + mehg-uptake-risk

;; -----
;; Track cumulative raw & normalized exposure
;; -----
set mehg-exposure-total mehg-exposure-total + MeHg-current
set mehg-exposure-total-normalized mehg-exposure-total-normalized + MeHg-normalized

;; -----
;; Patch-level risk recording (species specific)
;; -----
let patch-risk mehg-uptake-risk

```

```
if breed = alewives [  
  ask patch-here [  
    set MeHg-exp-risk-alewife      patch-risk  
    set MeHg-exp-risk-total-alewife (MeHg-exp-risk-total-alewife + patch-risk)  
  ]  
]  
  
if breed = stripedbass [  
  ask patch-here [  
    set MeHg-exp-risk-stripedbass  patch-risk  
    set MeHg-exp-risk-total-stripedbass (MeHg-exp-risk-total-stripedbass + patch-risk)  
  ]  
]  
end
```


Chapter 9

Small Particulate Feeding

9.1 Overview

Filter feeding is a foraging behavior in which fish consume suspended particulate matter (SPM) while moving through estuarine and coastal systems, linking food availability directly to hydrodynamic transport processes. In this submodel, agents evaluate neighboring patches based on SPM concentration and selectively feed where energetic return is highest, following optimal foraging principles that maximize net energy gain relative to the costs of searching, handling, and movement (?). Filtration capacity scales with body size, metabolic demand, and digestion efficiency, consistent with bioenergetic constraints on intake and processing capacity (?), and is reduced under competition from other filter-feeding fish.

Within this submodel, SPM includes both inorganic particles and organic material such as algae, detritus, and zooplankton, acting as both a food resource and a vector for contaminant transport in estuarine systems (??). As a result, contaminant exposure through this pathway is directly linked to feeding behavior, where assimilated biomass carries associated mercury and methylmercury that are incorporated into internal stores through biomagnification processes (?). This formulation allows feeding-driven exposure to emerge from the interaction between hydrodynamic SPM fields, foraging decisions, and physiological constraints, rather than being treated as a static uptake rate.

9.2 Purpose and Patterns

The purpose of this submodel is to simulate ecologically realistic filter feeding behavior in migratory fish using suspended particulate matter as a food resource. This behavior links hydrodynamic particle fields to energy gain, contaminant exposure, and movement decisions. Filter feeding only occurs when spatial

conditions and energetic returns are favorable, and incorporates competition, biomagnification, and digestion constraints.

This submodel represents the following ecological patterns:

- **Filtration capacity increases with body size, metabolism, and digestive efficiency**
Larger individuals with higher metabolism filter more water and assimilate more prey mass.
- **Gaussian SPM preference centered on optimal SPM**
Feeding intensity is highest near an optimal SPM range and decreases away from that peak.
- **Crowding penalty from con-specifics**
Filtration benefit decreases when multiple individuals feed in the same patch.
- **Movement toward highest benefit**
Fish move toward the patch with the highest filtration benefit based on SPM and competitive pressure.
- **Contaminant assimilation tied directly to filtration**
Mercury and methylmercury are assimilated proportionally to SPM concentration and biomagnification factors derived from empirical trophic magnification relationships.

9.3 Entities, State Variables, and Scales

9.3.1 Spatial and Temporal Scales

Spatial Unit: Patch (3 m x 3 m resolution)

Temporal Unit: 5 minute time steps (tick)

9.3.2 Global Variables

Global Variable	Definition
spm-sd	Standard deviation for Gaussian SPM preference curve (controls width of feeding preference)
max-filter-distance	Maximum distance within which patches can be evaluated (optional, model dependent)
BMF-MeHg	Biomagnification factor for methylmercury (1.74; after Lavoie et al. 2013)
BMF-Hg	Biomagnification factor for inorganic Hg (1.38; after Lavoie et al. 2013)

Global Variable	Definition
previous-x	X-coordinate of agent's previous location (used to compute travel distance and energy cost)
previous-y	Y-coordinate of agent's previous location (used with previous-x)

9.3.3 Patch Variables

Variable Name	Definition
SPM	Suspended particulate matter concentration in the water column
mercury	Inorganic mercury concentration at the patch
methylmercury	Methylmercury concentration at the patch
patch-terrain	Identifies water or land patches

9.3.4 Agent Variables

Variable Name	Definition
size	Body size used to scale filtration capacity
weight	Biomass used in base filtration scaling (from metabolism submodel)
metabolism-rate	Metabolic rate that modulates filtration intensity
digestion-efficiency	Fraction of filtered biomass digested
optimal-SPM	SPM concentration where filtration is most efficient
filter-feed?	Boolean indicating whether the agent engages in filter feeding
energy	Current energy reserves
E-swim	Energy cost of swimming per meter
swim-efficiency	Efficiency of movement based on metabolic state
speed	Swimming speed
stomach-contents	Biomass currently held in the stomach
stomach-capacity	Maximum biomass that can be held
stomach-contents-Hg	Inorganic mercury within stomach contents

Variable Name	Definition
stomach-contents-MeHg	Methylmercury within stomach contents
hg-total	Total cumulative inorganic mercury body burden
mehg-total	Total cumulative methylmercury body burden
hg-exposure-total	Cumulative Hg exposure from all pathways
mehg-exposure-total	Cumulative MeHg exposure from all pathways
hg-exposure-total-normalized	Normalized inorganic mercury exposure
mehg-exposure-total-normalized	Normalized methylmercury exposure
previous-x (<i>if not global</i>)	Previous x-location for travel-distance calculation
previous-y (<i>if not global</i>)	Previous y-location

9.4 Process Overview and Scheduling

1. Compute filtration capacity based on size, metabolism, and digestion efficiency.
2. Evaluate neighboring patches using a Gaussian suitability curve around optimal SPM, reflecting unimodal feeding performance where filtration efficiency and energetic return peak at intermediate resource concentrations and decline at both low and high extremes due to limitation or processing constraints (?).
3. Apply a competition penalty based on the number of conspecifics feeding on each patch to represent density-dependent foraging, where increased local competition reduces per-capita resource availability and filtration efficiency, consistent with optimal foraging predictions under shared resource use (?) and observed density-dependent feeding dynamics in SPM-mediated systems (?).
4. Select the patch with the highest filtration benefit.
5. Move toward the chosen patch with travel distance reduced by swimming efficiency.
6. Apply swimming energy cost (energy units).
7. Filter feed by assimilating SPM scaled biomass from the chosen patch.
8. Assimilate mercury and methylmercury using biomagnification factors from (?).
9. Add assimilated biomass (g) and contaminants (ng) to stomach contents .

9.5 Design Concepts

Basic Principles: Small particulate feeding reflects how pelagic and estuarine fish forage on suspended particles while migrating (?). Filtration scales with size and metabolic demand consistent with bioenergetic scaling principles (?). Suspended particulate matter (SPM) influences both food availability and contaminant exposure pathways in estuarine systems (???)

Emergence: Feeding hotspots emerge where hydrodynamic SPM fields align with optimal SPM and competition is minimal. Contaminant loads emerge from local mercury and methylmercury concentrations and biomagnification.

Adaptation: Agents adjust their direction and feeding behavior based on SPM quality, filtration capacity, and local competition.

Sensing: Agents sense SPM, mercury, methylmercury, competition from conspecifics, stomach fullness, and their own metabolic and digestive state.

Stochasticity: Variation in SPM quality, contaminant concentrations, and local competition create variability in feeding and assimilation outcomes.

Interaction: Competition reduces filtration benefit when multiple conspecifics occupy the same patch, consistent with density-dependent foraging dynamics in spatially structured systems (?).

Observation: The model tracks biomass intake, mercury and methylmercury assimilation, stomach contents, and contaminant load added per patch.

9.6 Initialization

Variable	Initialized Value	Justification
stomach-contents	0	No biomass in the stomach at model start
stomach-contents-Hg	0	No initial inorganic mercury from feeding
stomach-contents-MeHg	0	No initial methylmercury
digestion-efficiency	Species specific	Determines fraction of filtered biomass assimilated
optimal-SPM	Species specific	Represents preferred SPM concentration

9.7 Submodels

9.7.1 Filtration Capacity

Filtration scales with size, metabolism, and digestion efficiency, consistent with bioenergetic constraints derived largely from studies of ectothermic aquatic organisms, particularly fishes (?):

$$Filter_{base} = \left(\frac{size}{100} \right) \cdot Met_{tick} \cdot Eff_{digest} \quad (C.46)$$

Where:

- Met_{tick} is the total metabolic rate for the current simulation tick ((energy units) tick⁻¹).
- Eff_{digest} is efficiency factor controlling digestion and filtration (%). User-defined value.
- $Filter_{base}$ is the baseline filtration capacity for SPM feeding (g tick⁻¹).

9.7.2 SPM Interest (Gaussian Suitability)

Suitability peaks at the optimal SPM concentration and declines with deviation from that optimum following a Gaussian performance curve (?), capturing a unimodal response to environmental gradients, representing maximal feeding performance at intermediate SPM and reduced suitability at both lower and higher concentrations:

$$A_{interest} = e^{-\frac{(SPM - SPM_{optimal})^2}{2\sigma^2}} \quad (C.47)$$

Where:

- $A_{interest}$ is the suitability weight for SPM-based feeding (dimensionless).
- $SPM_{optimal}$ is the optimal SPM concentration for feeding suitability ($\frac{kg}{m^3}$).
- SPM is the suspended particulate matter (SPM) concentration on the patch ($\frac{kg}{m^3}$).
- σ is the standard deviation tolerance for suitability curve ($\frac{kg}{m^3}$).

9.7.3 Resource Benefit Calculation

Benefit for each neighboring patch:

$$A_{benefit} = SPM \cdot A_{interest} \cdot Filter_{base} \quad (C.48)$$

Where:

- $A_{benefit}$ is the relative feeding benefit score for a candidate patch (dimensionless).

- $A_{interest}$ is the suitability weight for SPM-based feeding (dimensionless).
- SPM is the suspended particulate matter (SPM) concentration on the patch ($\frac{kg}{m^3}$).
- $Filter_{base}$ is the baseline filtration capacity for SPM feeding (g tick⁻¹).

If con-specifics are present:

$$A_{benefit} = \frac{A_{benefit}}{N_{conspecifics}} \quad (C.49)$$

Where:

- $A_{interest}$ is the suitability weight for SPM-based feeding (dimensionless).
- $N_{conspecifics}$ is the number of conspecifics on a patch (agents).

9.7.4 Movement

Travel distance is reduced by swimming efficiency:

$$D_{travel} = \min(D_{to-patch}, A_{speed} \cdot swim_{Eff}) \quad (C.50)$$

Where:

- D_{travel} is the distance traveled in the current tick (patch units).
- $D_{to-patch}$ is the distance to a candidate target patch (patch units).
- A_{speed} is the actual swimming speed used for movement (m tick⁻¹).
- $swim_{Eff}$ is the efficiency multiplier applied to swimming energetics (%).

Energy cost of swimming:

$$E_{agent} = E_{agent} - (E_{swim} \cdot D_{travel}) \quad (C.51)$$

Where:

- E_{agent} is the total internal energy available to the agent (energy units).
- E_{swim} is the energy cost of swimming for the tick ((energy units) patch⁻¹).
- D_{travel} is the distance traveled in the current tick (patch units).

9.7.5 Feeding and Stomach Loading

Assimilated biomass:

$$M_{intake} = \min(M_{prey}, stomach_{max} - M_{stomach}) \quad (C.52)$$

Where:

- M_{intake} is the biomass actually ingested by the agent (g).

- M_{prey} is the biomass available from SPM filtration (g).
- $stomach_{max}$ is an agent's maximum stomach capacity (g).
- $M_{stomach}$ is the biomass currently stored in the stomach (g).

9.7.6 Contaminant Assimilation and Biomagnification

Biomagnification factors follow ?:

$$BMF_{MeHg} = 10^{(0.24 \cdot \Delta TP)} \quad (C.53)$$

Where:

- $\Delta TP = 1$ is the trophic position difference for biomagnification.
- BMF_{MeHg} is the biomagnification factor for methylmercury (dimensionless).

$$BMF_{MeHg} = 10^{0.24} \approx 1.74 \quad (C.54)$$

Where:

- BMF_{MeHg} is the biomagnification factor for methylmercury (dimensionless).

Assimilation equations:

$$MeHg_{intake} = SPM \cdot MeHg_{patch} \cdot BMF_{MeHg} \cdot M_{intake} \quad (C.55)$$

Where:

- $MeHg_{intake}$ is the methylmercury biomass actually ingested by the agent (ng).
- SPM is the suspended particulate matter (SPM) concentration on the patch ($\frac{kg}{m^3}$).
- $MeHg_{patch}$ is the patch-level methylmercury used in feeding uptake ($ng\ g^{-1}$).
- BMF_{MeHg} is the biomagnification factor for methylmercury.
- M_{intake} is the biomass actually ingested by the agent (g).

Stomach contaminant update:

$$MeHg_{stomach} = MeHg_{stomach} + MeHg_{intake} \quad (C.56)$$

Where:

- $MeHg_{stomach}$ is the methylmercury currently stored in the stomach (ng).
- $MeHg_{intake}$ is the methylmercury biomass actually ingested by the agent (ng).

9.8 Netlogo Implementation

```

globals [
  ;; Filter feeding parameters
  spm-sd                ;; standard deviation for Gaussian SPM preference curve (25)
  max-filter-distance   ;; how far fish can evaluate patches (optional)

  ;; Biomagnification constants from Lavoie et al. 2013
  BMF-MeHg              ;; 1.74 (MeHg biomagnification factor)
  BMF-Hg                ;; 1.38 (Hg biomagnification factor)
]

patches-own [
  ;; physical environment
  SPM                  ;; suspended particulate matter
  patch-terrain        ;; "water" or "land"

  ;; contaminant concentrations
  mercury              ;; inorganic Hg
  methylmercury        ;; MeHg
]

fish-own [
  ;; -----
  ;; Filter feeding physiology
  ;; -----
  size                 ;; body size scaling filtration
  metabolism-rate      ;; modulates filtration intensity
  digestion-efficiency  ;; fraction of filtered biomass digested
  optimal-SPM          ;; preferred SPM concentration
  filter-feed?         ;; Boolean: whether agent filter feeds

  ;; -----
  ;; Movement & energy
  ;; -----
  speed                ;; movement rate
  swim-efficiency      ;; metabolism-adjusted efficiency
  E-swim               ;; energy cost per unit distance
  energy               ;; energetic budget

  ;; -----
  ;; Stomach contents
  ;; -----
  stomach-contents

```

```

stomach-capacity
stomach-contents-Hg          ;; inorganic Hg from filter feeding
stomach-contents-MeHg        ;; MeHg from filter feeding

;; -----
;; Contamination physiology
;; (needed for risk integration)
;; -----
hg-total                    ;; cumulative Hg body burden
mehg-total                  ;; cumulative MeHg body burden
hg-exposure-total
mehg-exposure-total
hg-exposure-total-normalized
mehg-exposure-total-normalized
]

to filter-feed
if filter-feed? [
  let spm-sd 25
  ;; Base filtration capacity scales with body size *and* metabolism
  let base-filtration ((weight / 100) * metabolism-rate)
  set base-filtration (base-filtration * digestion-efficiency)
  let neighbors-list sort (neighbors with [patch-terrain = "water"])
  let best-patch nobody
  let best-score -999

  foreach neighbors-list [p ->
    let spm-patch [SPM] of p
    let interest exp (- ((spm-patch - optimal-SPM) ^ 2) / (2 * (spm-sd ^ 2)))
    let benefit (spm-patch * interest * base-filtration)
    let comp count alewives-on p
    if comp > 0 [ set benefit benefit / comp ]
    if benefit > best-score [
      set best-score benefit
      set best-patch p
    ]
  ]
]

if best-patch != nobody and [patch-terrain] of best-patch = "water" [
  face best-patch
  face best-patch
  move-to best-patch

  ;; calculate distance between best patch and previous patch
  let travel-distance distancexy previous-x previous-y

```

```

;; --- Swimming energy cost (scaled by efficiency) ---
set energy max list 0 (energy - (E-swim * travel-distance))

;; --- Feeding and digestion (temperature-dependent) ---
let patch-spm [SPM] of best-patch
let prey-assimilated patch-spm * base-filtration * 300

;; --- Contaminant uptake (SPM × Hg and MeHg) ---
let spm-hg [mercury] of best-patch
let spm-mehg [methylmercury] of best-patch

;; --- Biomagnification ---
let biomag-factor (random-float 30) + 60 ;; random 60-90
let hg-assimilated (patch-spm * spm-mehg * biomag-factor * prey-assimilated)
let mehg-assimilated (patch-spm * spm-hg * (biomag-factor / 10) * prey-assimilated)

;; Remaining capacity
let space-available (stomach-capacity - stomach-contents)

;; You can only eat as much as fits
let intake (min (list prey-assimilated space-available))

;; Add biomass to stomach
set stomach-contents (stomach-contents + intake)
set stomach-contents-Hg (stomach-contents-Hg + hg-assimilated);; mercury stomach amount
set stomach-contents-MeHg (stomach-contents-MeHg + mehg-assimilated);; methylmercury stomach

]
end

```

9.9 Overview

Lipid catabolism provides a secondary energy pathway that fish use when immediate food resources are unavailable or when energy falls below a functional threshold. In this submodel, agents metabolize stored lipid reserves to restore energy, maintain migration, and avoid starvation, with lipid use influenced by body condition, metabolic efficiency, and the magnitude of energetic deficit. This process represents endogenous energy mobilization when metabolic demand exceeds energy intake, as energetic deficits generated by movement, digestion, and osmoregulation trigger the use of stored lipids to maintain function, consistent with established physiological responses in fishes during fasting, migration, and overwintering (?). In anadromous species, lipid reserves play a critical role in sustaining upstream migration and reproductive effort when feeding is reduced or suspended (?).

Lipid depletion reduces body weight over time, creating a biologically meaningful cost to sustained reliance on internal reserves and linking energy balance to physiological state. Contaminant uptake does not occur during lipid catabolism, reflecting the absence of feeding. This formulation allows energy limitation and recovery to emerge dynamically from the interaction between metabolic demand and resource availability.

9.10 Purpose

The purpose of this submodel is to simulate endogenous energy mobilization through lipid reserves when fish cannot meet energetic needs through feeding. Lipid catabolism allows migratory fish to continue movement and survive temporary periods of low food availability. This mechanism reflects empirical observations of lipid use during spawning migrations, overwintering, and other energetically demanding life history periods.

This submodel is best used in tandpm with an energy recovery strategy like filter feeding and digestion, or pref=ation and digesiton

9.11 Entities, State Variables, and Scales

9.11.1 Spatial and Temporal Scales

Spatial Unit: Patch (3 m x 3 m resolution)

Temporal Unit: 5 minute time steps (tick)

9.11.2 Global Variables

Global Variable	Definition
full-energy-level	Target energy level used to compute energetic deficit during lipid use (default = 100).
energy-per-lipid	Energy returned per unit lipid mass mobilized (default = 40), used to convert lipid into energy.
lipid-refill-fraction	Fraction of body mass available to mobilize as lipid during fasting (default = 0.5).
lipid-max-fraction (<i>optional</i>)	Upper bound on lipid fraction that can be lost before mortality or physiological collapse (not currently enforced but biologically relevant).
debug-lipid? (<i>optional</i>)	Toggles printing of intermediate lipid catabolism calculations for debugging.

9.11.3 Patch Variables

Lipid catabolism does not directly depend on patch level variables, since no external resources are consumed.

9.11.4 Agent Variables

Variable Name	Definition
energy	Current energy level of the agent
weight	Current body mass of the fish
original-weight	Initial body mass used to scale maximum lipid availability
lipid-catabolism-efficiency	Efficiency with which lipids are converted into usable energy. This value is equal to the current metabolic rate divided by the metabolic rate at optimal temperature, allowing lipid use to scale with physiological performance
lipid-used	Amount of lipid converted to energy during the current step
metabolism-rate	Current metabolic rate, which influences efficiency scaling
baseline-metabolism-rate	Metabolic rate at optimal temperature
mehg-foraging	Methylmercury acquired during feeding (set to zero when lipids are used)
hg-foraging	Inorganic mercury acquired during feeding (set to zero when lipids are used)
mehg-foraging-total	Total methylmercury obtained from foraging
hg-foraging-total	Total inorganic mercury obtained from foraging
mehg-total	Total methylmercury accumulated
hg-total	Total inorganic mercury accumulated

9.12 Process Overview and Scheduling

1. Compute energy deficit relative to a target energy level of one hundred (% relative density).
2. Determine the lipid amount needed to fill the deficit (g tick⁻¹).

3. Limit lipid use by the maximum lipid available for mobilization (g tick⁻¹).
4. Apply lipid catabolism efficiency to determine usable lipid energy (%).
5. Convert mobilized lipid (g) to energy (energy units).
6. Reduce weight (g) by the amount of lipid burned (g).

9.13 Design Concepts

Basic Principles: Lipid catabolism represents endogenous energy mobilization when external food intake is insufficient. Fish draw upon stored lipids under energetic deficit, a well documented physiological response during migration, overwintering, and fasting (?). In anadromous species, these lipid reserves fuel upstream migration and reproductive effort when feeding is reduced or suspended [?;].

Emergence: Patterns of weight loss, energy cycling, and recovery emerge from repeated lipid use during periods of limited food access.

Adaptation: Agents adaptively mobilize lipids based on the severity of their energy deficit and on efficiency scaling from the metabolism submodel.

Objectives: Agents seek to maintain enough energy to migrate, forage, avoid predation, and prevent starvation.

Sensing: Agents sense their own internal energy level and lipid reserve availability.

Interaction: Lipid catabolism does not require environmental interaction and therefore represents a purely endogenous process.

Stochasticity: This submodel is deterministic. However, variation in weight, metabolism, and previous feeding creates emergent differences among individuals.

Observation: Outputs include energy restored through lipid catabolism, lipid used, and body mass loss over time.

9.14 Initialization

Variable	Initialized Value	Justification
weight	species specific	Establishes initial body condition
original-weight	equal to initial weight	Defines total lipid availability
energy	100	Full energy at initialization

Variable	Initialized Value	Justification
lipid-catabolism-efficiency	from metabolism submodel	Reflects physiological scaling of lipid use
lipid-used	0	No lipid used at start

9.15 Submodels

9.15.1 Lipid Mobilization

Maximum lipid available for mobilization depends on relative body mass:

$$Lipid_{refill} = 0.5 \times \left(\frac{A_{mass}}{A_{mass_i}} \right) \quad (C.57)$$

Where:

- $Lipid_{refill}$ is the Maximum lipid available for mobilization this tick (g).
- A_{mass} is the agent's body mass used for allometric and energetic scaling (g).

Energy deficit relative to full energy:

$$E_{deficit} = 100 - E_{agent} \quad (C.58)$$

Where:

- $E_{deficit}$ is the energy shortfall relative to full energy (energy units).
- E_{agent} is the total internal energy available to the agent (energy units).

Lipid required to refill the deficit:

$$Lipid_{needed} = \frac{Energy_{deficit}}{100} \quad (C.59)$$

Where:

- $Lipid_{needed}$ is the lipid required to fill the energy deficit (g).
- $E_{deficit}$ is the energy shortfall relative to full energy (% , relative energy).

Actual lipid mobilized:

$$Lipid_{used} = \min(Lipid_{needed}, Lipid_{refill}) \times Lipid_{Eff} \quad (C.60)$$

Where:

- $Lipid_{used}$ is the lipid mobilized during the current tick (g).
- $Lipid_{needed}$ is the lipid required to fill the energy deficit (g).
- $Lipid_{refill}$ is the Maximum lipid available for mobilization this tick (g).
- $Lipid_{Eff}$ is the efficiency multiplier applied to lipid catabolism (%).

9.15.2 Energy Conversion

Mobilized lipid is converted into energy:

$$E_{added} = Lipid_{used} \times 100 \quad (C.61)$$

Where:

- E_{added} is the energy added from lipid mobilization (energy units).
- $Lipid_{used}$ is the lipid mobilized during the current tick (g).

Energy is capped at one hundred:

$$E_{agent} = \min(100, E_{agent} + E_{added}) \quad (C.62)$$

Where:

- E_{agent} is the total internal energy available to the agent (energy units).
- E_{added} is the energy added from lipid mobilization (energy units).

9.15.3 Weight Loss

Weight decreases proportionally to lipid used:

$$A_{mass} = A_{mass} - Lipid_{used} \quad (C.63)$$

Where:

- A_{mass} is an agent's body mass used for allometric and energetic scaling (g).
- $Lipid_{used}$ is the lipid mobilized during the current tick (g).

9.15.4 Contaminant Bookkeeping

No mercury or methylmercury is acquired during lipid catabolism:

$$MeHg_{digested} = 0$$

$$Hg_{digested} = 0$$

These values are added to cumulative totals to maintain bookkeeping consistency, but do not increase contaminant burden.

9.16 Netlogo Implementation

```

;; =====
;; LIPID CATABOLISM SUBMODEL DECLARATIONS
;; =====

turtles-own [

  ;; Energetics & body condition
  energy
  weight
  original-weight

  ;; Lipid metabolism
  lipid-catabolism-efficiency
  lipid-used

  ;; Metabolic rate inputs
  metabolism-rate
  baseline-metabolism-rate

  ;; Contaminant bookkeeping during fasting
  meh-g-foraging
  hg-foraging

  meh-g-foraging-total
  hg-foraging-total

  meh-g-total
  hg-total
]

;; =====
;; LIPID CATABOLISM SUBMODEL
;; =====

to lipid-loss
  ;; Maximum lipid metabolized this step, adjusted by efficiency
  let lipid-refill 0.5 * (weight / original-weight)
  ;print (word "lipid-refill: " lipid-refill)
  ;print (word "current energy: " energy)
  ;; Calculate deficit and lipid need
  let energy-deficit (100 - energy)
  ;print (word "energy-deficit: " energy-deficit)
  let lipid-needed (energy-deficit / 40)

```

```

;print (word "lipid-needed: " lipid-needed)

;; Actual lipid use depends on efficiency and availability
let lipid-used (min list lipid-needed lipid-refill)
set lipid-used (lipid-used * lipid-catabolism-efficiency)
;print (word "lipid-used: " lipid-used)

;; Convert lipid to energy (scaled by efficiency)
let energy-change (lipid-used * 40)
;print (word "energy added: " energy-change)
set energy (energy + energy-change)
;print (word "energy after lipid conversion: " energy)
if energy > 100 [
  set energy 100
  print "energy capped at 100"
]

;; Weight loss proportional to lipid burned
set weight weight - (lipid-used)
;print (word "weight after lipid loss: " weight)

;; When relying on lipids, Hg mercury foraging risk stops

;; contaminant bookkeeping
let digested-MeHg 0
let digested-Hg 0
set mehg-foraging digested-MeHg
set hg-foraging digested-Hg

set mehg-foraging-total mehg-foraging-total + digested-MeHg
set hg-foraging-total hg-foraging-total + digested-Hg
set mehg-total mehg-total + digested-MeHg
set hg-total hg-total + digested-Hg
end

```

Chapter 10

Landward Migration

10.1 Overview

This function simulates landward migration for fish moving upstream through riverine and estuarine systems, where agents adjust movement based on local water velocity, energetic condition, and the relative difficulty of swimming in a given flow. Within this submodel, upstream movement emerges from the interaction between hydrodynamic forcing, body size, and energetic state, as agents navigate spatially variable flow fields where velocity and depth constrain swimming difficulty and energetic cost. These relationships are consistent with allometric scaling in aquatic organisms (?) and observed behavioral responses of fish to hydrodynamic conditions during migration (?).

Movement is structured through a least-cost framework that is recalculated under changing hydrodynamic conditions, where agents respond locally to gradients in flow resistance and path difficulty rather than optimizing globally. This formulation allows migration pathways, travel time, and energy expenditure to emerge dynamically from repeated interactions between organismal capacity and the hydrodynamic environment.

10.2 Purpose

The purpose of this submodel is to represent upstream migration under realistic spatial hydrodynamic resistance. Agents evaluate flow velocity, movement difficulty, body condition, and patch level travel costs before advancing along a least cost path toward a home location.

10.3 Entities, State Variables, and Scales

10.3.1 Spatial and Temporal Scales

Spatial Unit: Patch (3 m x 3 m resolution)

Temporal Unit: 5 minute time steps (tick)

10.3.2 Global Variables

Variable	Definition
max-seaward-velocity	Maximum seaward directed velocity in the domain, used for normalization.
max-landward-velocity	Maximum landward directed velocity in the domain.

10.3.3 Patch Variables

Variable	Definition
velocity	Depth averaged hydrodynamic velocity at the patch.
depth	Water depth at the patch.
patch-terrain	Indicates whether the patch is water or land.
cost-to-home	Patch level cost field producing a least cost corridor.
visits-by-alewife	Number of times the agent visits the patch.
ticks-spent-alewife	Time spent on the patch.

10.3.4 Agent Variables

Variable	Definition
weight	Body mass of the fish.
speed	Current swimming speed.
prev-speed	Previous time step swimming speed.
max-speed	Maximum sustained speed.
min-speed	Minimum allowable speed.
swim-efficiency	Scaling factor for acceleration and deceleration.
difficulty-factor	Difficulty of movement under local velocity conditions.

Variable	Definition
planned-path	Sequence of patches in the least cost route.
home-patch	Destination patch for migration.
trail	Record of patches visited by the agent.

10.4 Process Overview and Scheduling

1. Compute swimming difficulty from hydrodynamic velocity and body size.
2. Calculate swimming speed using desired-speed, velocity influence, and acceleration limits.
3. Evaluate the least cost field and construct a path to the home patch.
4. Move along the path for a distance proportional to swimming speed.
5. Record patch visitation and time spent.

10.5 Design Concepts

Basic Principles: Hydrodynamic constraints, difficulty scaling, and body condition influence agent movement.

Emergence: Migration routes, temporal occupancy, and movement speed emerge from repeated interactions of flow and difficulty.

Objectives: Agents attempt to reach the home patch but do not optimize globally. Movement follows the least cost route at each step.

Sensing: Agents sense velocity and depth of the local and neighboring patches.

Observation: Paths, timing, and residency can be monitored chronologically.

10.6 Initialization

Variable	Initial Value	Justification
velocity	user-defined	Hydrodynamic input data.
weight	species-specific	Required for difficulty scaling.
max-speed	species-specific	Physiological constraint.
min-speed	species-specific	Prevents speed collapse.
swim-efficiency	species or constant	Controls acceleration rate.

Variable	Initial Value	Justification
difficulty-factor	1	Neutral starting difficulty.
prev-speed	starting speed	Smooth initial movement.

10.7 Submodels

10.7.1 Swimming Difficulty

Swimming difficulty reflects the hydrodynamic effort required for the agent to move through its current patch. Velocity is normalized between the maximum observed landward and seaward velocities:

$$V_{\text{normalized}} = \frac{V_{\text{patch}} - V_{\text{min}}}{V_{\text{max}} - V_{\text{min}}} \quad (\text{C.64})$$

Where:

- $V_{\text{normalized}}$ is the normalized patch velocity (dimensionless).
- V_{patch} is the depth-averaged along-channel water velocity at the patch (m s^{-1}).
- V_{min} is the minimum observed velocity (m s^{-1}).
- V_{max} is the maximum observed velocity (m s^{-1}).

This normalized velocity is scaled by relative body size $M_{\text{agent}}/M_{\text{max}}$ and raised to a scaling exponent k (??):

$$Df_{\text{raw}} = \left(\frac{V_{\text{normalized}}}{A_{\text{mass}}/M_{\text{max}}} \right)^k \quad (\text{C.65})$$

Where:

- Df_{raw} is the raw hydrodynamic difficulty prior to mapping (dimensionless).
- $V_{\text{normalized}}$ is the normalized patch velocity (m s^{-1}).
- A_{mass} is an agent's body mass used for allometric and energetic scaling (g).
- M_{max} is the maximum body mass across the population (g).
- k is the energy scaling exponent (dimensionless).

Difficulty is then mapped onto the one to ten range:

$$D_f = 1 + 9 \cdot Df_{\text{raw}} \quad (\text{C.66})$$

Where:

- D_f is the swimming difficulty index (dimensionless).
- Df_{raw} is the raw hydrodynamic difficulty prior to mapping (dimensionless).

and constrained:

$$D_f \in [1, 10] \quad (\text{C.67})$$

Where:

- D_f is the mapped swimming difficulty index (dimensionless, range 1–10).

Difficulty increases when:

- M_{agent} is small
- $|V_{\text{patch}}|$ approaches its maximum value
- flow opposes movement

10.7.2 Swimming Speed

Swimming speed depends on energetic condition, hydrodynamic difficulty, and the effect of local flow. The agent first computes an energy factor:

$$E_{\text{factor}} = \frac{E_{\text{agent}}}{100} \quad (\text{C.68})$$

Where:

- E_{factor} is the normalized energy level of the agent (dimensionless).
- E_{agent} is the total internal energy available to the agent (energy units).

Velocity impact from hydrodynamics is:

$$V_{\text{impact}} = k \cdot V_{\text{patch}} \cdot 300 \quad (\text{C.69})$$

Where:

- V_{impact} is the flow contribution to desired swimming speed (m tick⁻¹).
- k is the energy scaling exponent used in velocity, difficulty, and osmoregulation terms (dimensionless).
- V_{patch} is the depth-Averaged along-channel water velocity at the patch (m s⁻¹).
- 300 is the timestep conversion from seconds to one tick (s/tick, e.g., 5 minutes).

The desired swimming speed is:

$$V_{\text{desired}} = \frac{V_{\text{max}} \cdot E_{\text{factor}}}{D_f} + V_{\text{impact}} \quad (\text{C.70})$$

Where:

- $V_{desired}$ is the desired swimming speed before constraints (m tick⁻¹).
- V_{max} is the maximum observed velocity (m tick⁻¹).
- E_{factor} is the normalized energy level of the agent (dimensionless).
- D_f is the mapped swimming difficulty index (dimensionless).
- V_{impact} is the flow contribution to desired swimming speed (m tick⁻¹).

Speed is constrained between a minimum and maximum:

$$V_{desired} = \min(V_{max}, \max(V_{min}, V_{desired})) \quad (C.71)$$

Where:

- $V_{desired}$ is the desired swimming speed before constraints (m tick⁻¹).
- V_{max} is the maximum observed velocity (m tick⁻¹).
- V_{min} is the minimum observed velocity (m tick⁻¹).

Acceleration is smoothed using the previous speed:

$$\Delta V_{max} = 0.5 \cdot swim_{Eff} \quad (C.72)$$

Where:

- ΔV_{max} is the maximum allowable speed change per tick (m s⁻¹/tick).
- $swim_{Eff}$ is the efficiency multiplier applied to swimming energetics (%).

If $|V_{desired} - V_{prev}| > \Delta V_{max}$, where V_{prev} is the previous speed then speed moves gradually toward the desired value. Otherwise speed is set directly.

10.7.3 Path Based Movement

Landward migration uses a least cost routing strategy instead of direct vector movement. The number of patches traversed in a tick is based on swimming speed:

$$D_{travel} = \frac{A_{speed}}{3} \quad (C.73)$$

Where:

- D_{travel} is the distance traveled in the current tick (patch units), where 1 patch = 3 m.
- A_{speed} is the actual swimming speed used for movement (m tick⁻¹).

Movement follows:

1. compute cost-to-home
2. generate a path using decreasing cost

3. move through the first $\lceil \text{travel_distance} \rceil$ patches
4. record patch visits and time spent
5. append patch-here to the trail if new location

10.7.4 Travel Cost

Travel cost for a candidate patch incorporates velocity magnitude, slope, and difficulty:

$$\text{cost}_{\text{raw}} = |V_{\text{patch}}| + 0.5 \cdot \max(0, \Delta\text{depth}) \quad (\text{C.74})$$

Where:

- cost_{raw} is the base travel cost including velocity and slope (dimensionless).
- V_{patch} is the depth-Averaged along-channel water velocity at the patch (m s^{-1}).
- Δd is the change in depth between patches (m).

Difficulty multiplies the base cost:

$$\text{cost} = \text{cost}_{\text{raw}} \cdot D_f \quad (\text{C.75})$$

Where:

- cost is the difficulty-scaled travel cost per patch (energy units).
- cost_{raw} is the base travel cost including velocity and slope (dimensionless).
- D_f is the mapped swimming difficulty index (dimensionless).

This produces a cost-to-home field representing hydrodynamic resistance (dimensionless), which is used for route selection.

10.7.5 Swimming Energy

Swimming incurs a metabolic cost that scales with base swimming cost, difficulty, and swim efficiency.

Let $\beta = 0.75$.

Let the effective multiplier be:

$$M_{\text{swim}} = \frac{D_f}{\text{swim}_{\text{Eff}}} \quad (\text{C.76})$$

Where:

- M_{swim} is the swimming energy multiplier (dimensionless).
- D_f is the mapped swimming difficulty index (dimensionless).
- swim_{Eff} is the efficiency multiplier applied to swimming energetics (dimensionless).

Where lower efficiency results in higher energetic cost.

Energy expenditure is:

$$E_{\text{swim}} = \text{swim}_{\text{base}} \cdot M_{\text{swim}}^{\beta} \quad (\text{C.77})$$

Where:

- E_{swim} is the energy cost of swimming for the tick (energy units).
- $\text{swim}_{\text{base}}$ is the baseline energetic cost of swimming ((energy units) patch⁻¹).
- M_{swim} is the swimming energy multiplier (dimensionless).
- β is the exponent controlling swimming cost scaling (dimensionless).

Total energy is reduced accordingly:

$$E_{\text{agent}} = \max(0, E_{\text{agent}} - E_{\text{swim}}) \quad (\text{C.78})$$

Where:

- E_{agent} is the total internal energy available to the agent (energy units).
- E_{swim} is the energy cost of swimming for the tick (energy units).

Higher difficulty and lower swim efficiency increase metabolic cost.

10.8 Netlogo Implementation

```
;; =====
;; LANDWARD MIGRATION + HYDRODYNAMIC SWIMMING SUBMODEL DECLARATIONS
;; =====

globals [

  ;; Velocity normalization bounds
  max-seaward-velocity
  max-landward-velocity
]

patches-own [

  ;; Environmental attributes
  velocity
  depth
  patch-terrain    ;; "water" or "land"

  ;; Least-cost routing fields
```

```

cost-to-home
cost-to-sea
cost-to-prey

;; Tracking fish patch-level behavior
visits-by-fish
ticks-spent-fish
]

turtles-own [

  ;; Body condition
  weight

  ;; Swimming behavior
  speed
  prev-speed
  max-speed
  min-speed
  swim-efficiency

  ;; Hydrodynamic resistance
  difficulty-factor

  ;; Energetics
  E-swim
  swim-base

  ;; Migration targets
  home-patch
  migration-patch

  ;; Movement memory
  planned-path
  trail
  previous-patch
  previous-x
  previous-y
]

;; =====
;;  SWIMMING DIFFICULTY (LANDWARD)
;;  =====

```

```

to calculate-difficulty-landward

  let M-max max [weight] of breed      ;; largest fish in population
  let M-agent weight                    ;; focal individual
  let V-max max-seaward-velocity
  let V-min max-landward-velocity
  let k 0.75                           ;; scaling exponent

  ;; Normalize velocity
  let normalized-velocity (velocity - V-min) / (V-max - V-min)

  ;; Difficulty from normalized velocity + size ratio
  let Df-raw (normalized-velocity / (M-agent / M-max)) ^ k

  ;; Scale to range 1-10
  set difficulty-factor (1 + 9 * Df-raw)
  set difficulty-factor max list 1 (min list 10 difficulty-factor)
end

;; =====
;;  SWIMMING SPEED (LANDWARD)
;;  =====

to calculate-swimming-speed-landward

  let k -0.75                           ;; velocity influence
  let energy-factor (energy / 100)      ;; 0-1 scale
  let velocity-impact (k * velocity * 300) ;; 5-min timestep scaling

  ;; Preliminary target speed
  let desired-speed
    (max-speed * energy-factor / difficulty-factor)
    + velocity-impact

  ;; Clamp desired speed to biologically realistic bounds
  set desired-speed min list max-speed (max list min-speed desired-speed)

  ;; Smooth acceleration / deceleration
  let max-speed-change 0.5
  set max-speed-change (max-speed-change * swim-efficiency)

  if abs(desired-speed - prev-speed) > max-speed-change [
    if desired-speed > prev-speed [
      set speed prev-speed + max-speed-change

```



```

    ]
    if desired-speed < prev-speed [
      set speed prev-speed - max-speed-change
    ]
  ]

  if abs(desired-speed - prev-speed) <= max-speed-change [
    set speed desired-speed
  ]

  set prev-speed speed
end

;; =====
;; LEAST-COST ROUTING TO HOME (LANDWARD)
;; =====

to calculate-cost-to-home [homing-patch]

  ;; initialize routing fields
  ask home-patch [ set cost-to-home 0 ]
  ask patch-here [ set cost-to-home 1e9 ]

  let frontier (list home-patch)
  let df difficulty-factor          ;; hydrodynamic penalty

  while [not empty? frontier] [
    let current first frontier
    set frontier but-first frontier

    ask current [
      let current-cost cost-to-home

      ask neighbors4 with [patch-terrain = "water"] [

        let travel-cost compute-travel-cost current self df
        let new-cost current-cost + travel-cost

        if new-cost < cost-to-home [
          set cost-to-home new-cost
          set frontier lput self frontier
        ]
      ]
    ]
  ]
]

```

```

]
end

;; =====
;; BUILD OPTIMAL PATH ENTIRELY FROM COST-FIELD
;; =====

to-report build-path [current-patch homing-patch]

  let path (list current-patch)
  let cur current-patch
  let max-steps 2000

  while [cur != homing-patch and max-steps > 0] [
    set max-steps max-steps - 1

    let candidates [neighbors4] of cur
    set candidates candidates with [
      patch-terrain = "water" and
      cost-to-home < [cost-to-home] of cur
    ]

    if not any? candidates [
      report path
    ]

    let next-patch min-one-of candidates [cost-to-home]
    set path lput next-patch path
    set cur next-patch
  ]

  report path
end

;; =====
;; TRAVEL COST UNDER HYDRODYNAMIC RESISTANCE
;; =====

to-report compute-travel-cost [from-patch to-patch df]

  ;; velocity resistance
  let raw-vel [velocity] of to-patch
  let velocity-resistance

```

```

    ifelse-value is-number? raw-vel [abs raw-vel] [0]

    ;; depth slope component
    let from-level [depth] of from-patch
    let to-level   [depth] of to-patch
    let slope 0

    if (is-number? from-level and is-number? to-level) [
      set slope max list 0 (to-level - from-level)
    ]

    ;; hydrodynamic cost + slope penalty
    let base-cost velocity-resistance + (0.5 * slope)

    ;; difficulty multiplier
    let df2 ifelse-value is-number? df [df] [1]

    report max list 0 (base-cost * df2)
  end

;; =====
;; LANDWARD MIGRATION - PATH EXECUTION
;; =====

to migrate-landward

  let travel-distance (speed / 3)
  let path build-path patch-here home-patch
  set planned-path path

  if not empty? path [

    let steps-to-move ceiling travel-distance
    let move-patches sublist path 0 (min list steps-to-move length path)

    let old-patch patch-here

    foreach move-patches [ p ->

      ;; Record patch visitation
      if p != old-patch [
        ask p [ set visits-by-alewife visits-by-alewife + 1 ]
        set old-patch p
      ]
    ]
  ]

```

```

;; Move into patch
face p
move-to p

ask p [
  set ticks-spent-alewife ticks-spent-alewife + 1
]
]

;; Add patch to historical trail
if not member? patch-here trail [
  set trail lput patch-here trail
]
end

;; =====
;; UNIVERSAL SWIMMING ENERGY COST
;; =====

to calculate-swim-energy

  let beta 0.75

  let energy-multiplier
    (difficulty-factor * (1 / swim-efficiency))

  set E-swim
    (swim-base * (energy-multiplier ^ beta))

  set energy max list 0 (energy - E-swim)
end

```

Chapter 11

Seaward Migration

11.1 Overview

This function simulates seaward migration for fish moving downstream through riverine and estuarine systems. Agents respond to hydrodynamic conditions by adjusting swimming difficulty, swimming speed, and movement along a least cost pathway toward a downstream migration patch.

This submodel represents downstream movement following the same mechanistic framework as landward migration, where movement emerges from the interaction between hydrodynamic forcing, body size, and energetic state (??). In this case, movement occurs in the seaward direction, requiring a change in sign within velocity-based scaling terms to reflect alignment with downstream flow. As in landward migration, agents follow a least-cost pathway based on local hydrodynamic resistance, allowing downstream trajectories, travel time, and energy expenditure to emerge dynamically from the interaction between organismal capacity and the flow field.

11.2 Purpose

The purpose of this submodel is to represent downstream migration under realistic hydrodynamic resistance. Agents evaluate flow velocity, movement difficulty, energetic condition, and travel costs before advancing along a least cost route toward a migration destination.

11.3 Entities, State Variables, and Scales

11.3.1 Spatial and Temporal Scales

Spatial Unit: Patch (3 m x 3 m resolution)

Temporal Unit: 5 minute time steps (tick)

11.3.2 Global Variables

Variable	Definition
max-seaward-velocity	Maximum observed seaward velocity in the domain.
max-landward-velocity	Maximum observed landward velocity in the domain.

11.3.3 Patch Variables

Variable	Definition
velocity	Depth averaged hydrodynamic velocity.
depth	Water depth at the patch.
patch-terrain	Indicates whether the patch is land or water.
cost-to-sea	Patch-level travel cost for seaward movement.
visits-by-alewife	Number of times the agent visits the patch.
ticks-spent-alewife	Time spent on the patch.

11.3.4 Agent Variables

Variable	Definition
weight	Body mass of the fish.
speed	Current swimming speed.
prev-speed	Previous time step swimming speed.
max-speed	Maximum sustained speed.
min-speed	Minimum allowable speed.
swim-efficiency	Swimming acceleration and energy scaling.
difficulty-factor	Hydrodynamic difficulty.
planned-path	Sequence of patches in least cost route.
migration-patch	Destination patch.
trail	Record of visited patches.

11.4 Process Overview and Scheduling

1. Calculate swimming difficulty using normalized velocity and body size.

2. Update swimming speed using energy reserves, difficulty, and flow velocity.
3. Calculate cost-to-sea and construct least cost path.
4. Move along the first part of the path based on swimming speed.
5. Record patch visits and time spent.

11.5 Design Concepts

Basic Principles: Hydrodynamic forcing, body size, and energy availability determine downstream movement.

Emergence: Route choice, travel duration, and energetic depletion emerge from repeated state updates.

Objectives: Agents attempt to reach the seaward migration patch following a hydrodynamically efficient route.

Sensing: Agents sense local velocity, slope, difficulty, and patch cost gradients.

Observation: Movement trajectories, energy decline, and convergence routes can be monitored.

11.6 Initialization

Variable	Initial Value	Justification
velocity	Hydrodynamic input	Required environmental forcing.
weight	Species-specific	Required for difficulty scaling.
max-speed	Species-specific	Physiological constraint.
min-speed	Species-specific	Prevents collapse to zero speed.
swim-efficiency	Species or constant	Controls acceleration and deceleration.
difficulty-factor	1	Neutral start value.
prev-speed	Starting speed	Smooth initial movement.

11.7 Submodels

11.7.1 Swimming Difficulty

Velocity is normalized between minimum and maximum observed values:

$$V_{normalized} = \frac{V_{patch} - V_{min}}{V_{max} - V_{min}} \quad (C.79)$$

Where:

- $V_{normalized}$ is the normalized patch velocity (dimensionless).
- V_{patch} is the depth-averaged along-channel water velocity at the patch (m s^{-1}).
- V_{min} is the minimum observed velocity (m s^{-1}).
- V_{max} is the maximum observed velocity (m s^{-1}).

Difficulty scales with body size (??):

$$Df_{raw} = \left(\frac{V_{normalized}}{A_{mass}/M_{max}} \right)^k \quad (\text{C.80})$$

Where:

- Df_{raw} is the raw hydrodynamic difficulty prior to mapping (dimensionless).
- $V_{normalized}$ is the normalized patch velocity (dimensionless).
- A_{mass} is an agent's body mass used for allometric and energetic scaling (g).
- M_{max} is the maximum body mass across the population (g).
- k is the energy scaling exponent (dimensionless).

Difficulty is mapped onto the one-to-ten range:

$$D_f = 1 + 9 \cdot Df_{raw} \quad (\text{C.81})$$

Where:

- D_f is the mapped swimming difficulty index (dimensionless).
- Df_{raw} is the raw hydrodynamic difficulty prior to mapping (dimensionless).

and constrained:

$$D_f \in [1, 10] \quad (\text{C.82})$$

Where:

- D_f is the mapped swimming difficulty index (dimensionless, range 1-10).

11.7.2 Swimming Speed

Energy factor:

$$E_{\text{factor}} = \frac{E_{\text{agent}}}{100} \quad (\text{C.83})$$

Where:

- E_{factor} is the normalized energy level of the agent (dimensionless).
- E_{agent} is the total internal energy available to the agent (energy units).

Velocity impact from hydrodynamics is:

$$V_{\text{impact}} = k \cdot V_{\text{patch}} \cdot 300 \quad (\text{C.69})$$

Where:

- V_{impact} is the flow contribution to desired swimming speed (m tick⁻¹).
- k is the energy scaling exponent used in velocity, difficulty, and osmoregulation terms (dimensionless).
- V_{patch} is the depth-Averaged along-channel water velocity at the patch (m s⁻¹).
- 300 is the timestep conversion from seconds to one tick (s/tick, e.g., 5 minutes).

The desired swimming speed is:

$$V_{\text{desired}} = \frac{V_{\text{max}} \cdot E_{\text{factor}}}{D_f} + V_{\text{impact}} \quad (\text{C.70})$$

Where:

- V_{desired} is the desired swimming speed before constraints (m tick⁻¹).
- V_{max} is the maximum observed velocity (m tick⁻¹).
- E_{factor} is the normalized energy level of the agent (dimensionless).
- D_f is the mapped swimming difficulty index (dimensionless).
- V_{impact} is the flow contribution to desired swimming speed (m tick⁻¹).

Speed is constrained between a minimum and maximum:

$$V_{\text{desired}} = \min(V_{\text{max}}, \max(V_{\text{min}}, V_{\text{desired}})) \quad (\text{C.71})$$

Where:

- V_{desired} is the desired swimming speed before constraints (m tick⁻¹).
- V_{max} is the maximum observed velocity (m tick⁻¹).
- V_{min} is the minimum observed velocity (m tick⁻¹).

Acceleration smoothing:

$$\Delta V_{\max} = 0.5 \cdot swim_{Eff} \quad (C.87)$$

Where:

- $\Delta V_{\max}$ is the maximum allowable speed change per tick (m tick⁻¹).
- $swim_{Eff}$ is the efficiency multiplier applied to swimming energetics (%).

If $|V_{\text{desired}} - V_{\text{prev}}| > \Delta V_{\max}$, where V_{prev} is the previous speed then speed moves gradually toward the desired value. Otherwise speed is set directly.

11.7.3 Path-Based Movement

Travel distance:

$$D_{\text{travel}} = \frac{A_{\text{speed}}}{3} \quad (C.88)$$

Where:

- D_{travel} is the distance traveled in the current tick (patch units), where 1 patch = 3 m.
- A_{speed} is the actual swimming speed used for movement (m tick⁻¹).

Movement follows:

1. compute cost-to-sea
2. build path using decreasing cost
3. traverse first $\lceil D_{\text{travel}} \rceil$ patches
4. record visitation and time spent
5. append to the movement trail

11.7.4 Travel Cost

Raw cost:

$$\text{cost}_{\text{raw}} = |V_{\text{patch}}| + 0.5 \cdot \max(0, \Delta d) \quad (C.89)$$

Where:

- cost_{raw} is the base travel cost including velocity and slope (dimensionless).
- V_{patch} is the depth-averaged along-channel water velocity at the patch (m s⁻¹).
- Δd is the change in depth between patches (m).

Difficulty scaling:

$$\text{cost} = \text{cost}_{\text{raw}} \cdot D_f \quad (\text{C.90})$$

Where:

- cost is the difficulty-scaled travel cost per patch (energy units).
- cost_{raw} is the base travel cost including velocity and slope (dimensionless).
- D_f is the mapped swimming difficulty index (dimensionless).

This produces a cost-to-sea field representing hydrodynamic resistance (dimensionless), which is used for route selection.

11.7.5 Swimming Energy

Let $\beta = 0.75$.

Energy multiplier:

$$M_{\text{swim}} = \frac{D_f}{\text{swim}_{\text{Eff}}} \quad (\text{C.91})$$

Where:

- M_{swim} is the swimming energy multiplier (dimensionless).
- D_f is the mapped swimming difficulty index (dimensionless).
- swim_{Eff} is the efficiency multiplier applied to swimming energetics (%).

Energy cost:

$$E_{\text{swim}} = \text{swim}_{\text{base}} \cdot M_{\text{swim}}^\beta \quad (\text{C.92})$$

Where:

- E_{swim} is the energy cost of swimming for the tick (energy units).
- $\text{swim}_{\text{base}}$ is the baseline energetic cost of swimming ((energy units) patch⁻¹).
- M_{swim} is the swimming energy multiplier (dimensionless).
- β is the exponent controlling swimming cost scaling (dimensionless).

Energy update:

$$E_{\text{agent}} = \max(0, E_{\text{agent}} - E_{\text{swim}}) \quad (\text{C.93})$$

Where:

- E_{factor} is the normalized energy level of the agent (energy units).
- E_{agent} is the total internal energy available to the agent (energy units).

11.8 NetLogo Implementation

```
;; =====
;; LANDWARD MIGRATION + HYDRODYNAMIC SWIMMING SUBMODEL DECLARATIONS
;; =====

globals [

  ;; Velocity normalization bounds
  max-seaward-velocity
  max-landward-velocity
]

patches-own [

  ;; Environmental attributes
  velocity
  depth
  patch-terrain    ;; "water" or "land"

  ;; Least-cost routing fields
  cost-to-home
  cost-to-sea
  cost-to-prey

  ;; Tracking fish patch-level behavior
  visits-by-fish
  ticks-spent-fish
]

turtles-own [

  ;; Body condition
  weight

  ;; Swimming behavior
  speed
  prev-speed
  max-speed
  min-speed
  swim-efficiency
]
```

```

;; Hydrodynamic resistance
difficulty-factor

;; Energetics
E-swim
swim-base

;; Migration targets
home-patch
migration-patch

;; Movement memory
planned-path
trail
previous-patch
previous-x
previous-y
]

;; =====
;; SEAWARD MIGRATION + HYDRODYNAMIC SWIMMING SUBMODEL
;; =====

;; -----
;; Swimming Difficulty (Seaward)
;; -----
to calculate-difficulty-seaward

  let M-max max [weight] of breed
  let M-agent weight
  let V-max max-landward-velocity
  let V-min max-seaward-velocity
  let k 0.75

  let normalized-velocity (velocity - V-min) / (V-max - V-min)

  let Df-raw (normalized-velocity / (M-agent / M-max)) ^ k

  set difficulty-factor (1 + 9 * Df-raw)
  set difficulty-factor max list 1 (min list 10 difficulty-factor)
end

;; -----
;; Swimming Speed (Seaward)

```

```
;; -----
to calculate-swimming-speed-seaward

  let k 0.75
  let energy-factor (energy / 100)

  let velocity-impact (k * velocity * 300)

  let desired-speed
    (max-speed * energy-factor / difficulty-factor)
    + velocity-impact

  set desired-speed
    min list max-speed (max list min-speed desired-speed)

  let max-speed-change 0.5
  set max-speed-change (max-speed-change * swim-efficiency)

  if abs(desired-speed - prev-speed) > max-speed-change [
    if desired-speed > prev-speed [
      set speed prev-speed + max-speed-change
    ]
    if desired-speed < prev-speed [
      set speed prev-speed - max-speed-change
    ]
  ]

  if abs(desired-speed - prev-speed) <= max-speed-change [
    set speed desired-speed
  ]

  set prev-speed speed
end
```

```
;; -----
;; Cost-to-Sea (Least Cost Routing)
;; -----
to calculate-cost-to-sea [sea-patch]

  ask migration-patch [ set cost-to-sea 0 ]
  ask patch-here      [ set cost-to-sea 1e9 ]

  let frontier (list migration-patch)
  let df difficulty-factor
```

```

while [not empty? frontier] [

  let current first frontier
  set frontier but-first frontier

  ask current [

    let current-cost cost-to-sea

    ask neighbors4 with [patch-terrain = "water"] [

      let travel-cost compute-travel-cost current self df
      let new-cost current-cost + travel-cost

      if new-cost < cost-to-sea [
        set cost-to-sea new-cost
        set frontier lput self frontier
      ]
    ]
  ]
]
end

```

```

;; -----
;; Build Path to Sea
;; -----
to-report build-path-sea [current-patch sea-patch]

  let path (list current-patch)
  let cur current-patch
  let max-steps 2000

  while [cur != migration-patch and max-steps > 0] [

    set max-steps max-steps - 1

    let candidates [neighbors] of cur
    set candidates candidates with [
      patch-terrain = "water" and
      cost-to-sea < [cost-to-sea] of cur
    ]

    if not any? candidates [
      report path
    ]
  ]

```

```

]

let next-patch min-one-of candidates [cost-to-sea]
set path lput next-patch path
set cur next-patch
]

report path
end

;; -----
;; Seaward Migration - Movement Execution
;; -----
to migrate-seaward

;; Schooling logic (excluded in documentation)
find-schoolmates
if any? schoolmates [
  find-nearest-neighbor
  ifelse distance nearest-neighbor < minimum-separation [
    separate
  ] [
    cohere
    align
  ]
  adjust-speed
]

;; Path-based movement
let travel-distance (speed / 3)
let path build-path-sea patch-here migration-patch
set planned-path path

if not empty? path [

  let steps-to-move ceiling travel-distance
  let move-patches sublist path 0 (min list steps-to-move length path)

  let old-patch patch-here

  foreach move-patches [ p ->

    if p != old-patch [
      ask p [ set visits-by-alewife visits-by-alewife + 1 ]
    ]
  ]
]

```



```
    set old-patch p
  ]

  face p
  move-to p

  ask p [
    set ticks-spent-alewife ticks-spent-alewife + 1
  ]
]

if not member? patch-here trail [
  set trail lput patch-here trail
]
end

;; -----
;;  Universal Swimming Energy Cost
;; -----
to calculate-swim-energy

  let beta 0.75
  let energy-multiplier (difficulty-factor * (1 / swim-efficiency))

  set E-swim (swim-base * (energy-multiplier ^ beta))
  set energy max list 0 (energy - E-swim)
end
```


Chapter 12

Schooling

12.1 Overview

Schooling is an emergent collective behavior in which fish align, cohere, and maintain spacing with nearby conspecifics, increasing hydrodynamic efficiency, predator avoidance, and coordinated migration. Within this model, schooling arises from local interactions among individuals, where agents adjust position, heading, and speed in response to nearby neighbors following the classic behavioral rules of separation, alignment, and cohesion. These rules are consistent with established models of collective fish behavior and agent-based schooling frameworks (??). As a result, group structure, synchronized movement, and cohesion emerge from decentralized decision-making rather than being imposed at the group level, allowing schooling dynamics to adapt to changing environmental conditions during migration.

The schooling behavior in this model is adapted from the publicly available NetLogo model Fish Schooling For Predator Avoidance created by Sebastian Dobon and shared through the NetLogo Modeling Commons. That model implements classic schooling rules from the original BOIDS framework developed by Craig Reynolds in 1987 in his paper *Flocks, Herds, and Schools: A Distributed Behavioral Model* (?).

12.2 Purpose

The purpose of this submodel is to simulate collective movement patterns observed in migratory fish where individuals adjust their position and speed based on nearby schoolmates. This mechanism enhances group cohesion during landward or seaward migration and produces emergent school structure, directional alignment, and spacing.

12.3 Entities, State Variables, and Scales

12.3.1 Spatial and Temporal Scales

Spatial Unit: Patch (3 m x 3 m resolution)

Temporal Unit: 5 minute time steps (tick)

12.3.2 Patch Variables

Variable Name	Definition
patch-terrain	Identifies water or land patches

12.3.3 Agent Variables

Variable Name	Definition
breed	Species identity used to restrict schooling to con-specifics
speed	Current swimming speed
min-speed	Lowest allowable swimming speed
start-migration?	Indicates whether the agent has begun migration
landward-migration?	Indicates landward migration mode
seaward-migration?	Indicates seaward migration mode
schoolmates	Set of nearby con-specifics that meet schooling criteria
nearest-neighbor	Closest schoolmate used for spacing and cohesion
minimum-separation	Desired minimum inter-fish spacing

12.4 Process Overview and Scheduling

1. Identify schoolmates based on species identity (categorical) and shared migration state (categorical).
2. Detect the nearest neighbor among schoolmates (distance, patch units, where 1 patch = 3 m).
3. Apply separation when neighbors are too close (threshold distance, patch units).
4. Apply cohesion and alignment when spacing is adequate (distance-based interaction, patch units).
5. Adjust speed to match group speed and maintain coordination (patch units tick⁻¹).

12.5 Design Concepts

Basic Principles: Schooling arises from local interaction rules of separation, cohesion, and alignment, implemented through the schooling and predator avoidance functions(?). These behavioral rules reflect well documented collective dynamics observed in migrating fishes, where individuals maintain group structure while balancing spacing, directional alignment, and coordinated movement.

Emergence: Cohesive schools, spacing patterns, and synchronized direction arise without centralized control.

Adaptation: Agents adjust heading and speed in response to nearby schoolmates.

Sensing

Agents sense species identity, distance to schoolmates, spacing, and local movement direction.

Interaction

Agents change speed and heading based on interactions with neighbors.

Stochasticity

Small random turning angles elsewhere in the model introduce variability in movement.

Collectives

Schools form as dynamic collectives with shared direction and spacing rules.

Observation

School size, spacing, direction, and cohesion may be tracked to evaluate emergent schooling patterns.

12.6 Initialization

Variable	Initialized Value	Justification
speed	species specific	Starting swimming speed
min-speed	species specific	Prevents schools from stopping entirely
schoolmates	empty set	No neighbors until movement begins
nearest-neighbor	none	Determined dynamically during the simulation
minimum-separation	species specific	Prevents collisions and maintains spacing

12.7 Submodels

12.7.1 Identifying Schoolmates

Fish select schoolmates from nearby con-specifics that share the same migration state, including the start of migration and direction (landward or seaward).

12.7.2 Nearest Neighbor

The closest schoolmate is used to determine whether spacing rules or cohesion rules should be applied.

12.7.3 Separation

If the nearest neighbor is closer than the minimum desired separation, the agent turns away to maintain spacing.

12.7.4 Cohesion

When spacing is appropriate, the agent turns toward the group centroid to maintain cohesive structure.

12.7.5 Alignment

The agent aligns its heading with the mean direction of nearby con-specifics to maintain synchronized movement.

12.7.6 Speed Adjustment

Speed is regulated to preserve cohesion within the school:

- The agent accelerates if any schoolmate is significantly faster.
- Otherwise, it gradually slows toward the minimum speed.

This produces emergent schooling speeds and prevents fragmentation.

12.8 Netlogo Implementation

```
;; -----
;; SCHOOLING SUBMODEL - VARIABLE DECLARATIONS
;; -----

patches-own [
  ;; Terrain classification
  patch-terrain      ;; "water" or "land"
]
```

```

turtles-own [
  ;; -----
  ;; Schooling state variables
  ;; -----
  schoolmates          ;; agentset of nearby con-specifics
  nearest-neighbor      ;; closest fish in schoolmates
  minimum-separation    ;; desired distance to avoid collisions

  ;; -----
  ;; Migration gating used for schooling eligibility
  ;; (schoolmates only form among agents sharing these states)
  ;; -----
  start-migration?      ;; has the agent initiated migration?
  landward-migration?   ;; migrating upstream?
  seaward-migration?    ;; migrating downstream?

  ;; -----
  ;; Swimming parameters affected by schooling
  ;; -----
  speed                 ;; current swimming speed
  min-speed              ;; lower speed limit to prevent stall
]

;; -----
;; SCHOOLING: MAIN CONTROLLER
;; -----

to school
  find-schoolmates

  if any? schoolmates [
    find-nearest-neighbor

    ;; Separation rule
    ifelse distance nearest-neighbor < minimum-separation
    [
      separate
    ]
    [
      ;; Cohesion + alignment rules
      cohere
      align
    ]
  ]

  ;; Adjust speed to maintain cohesion

```

```

    adjust-speed
  ]
end

;; -----
;;  FIND SCHOOLMATES (CON-SPECIFICS + SAME MIGRATION MODE)
;;  -----

to find-schoolmates
  set schoolmates other turtles with [
    breed = [breed] of myself and
    start-migration? and
    landward-migration? = [landward-migration?] of myself and
    seaward-migration? = [seaward-migration?] of myself
  ]
end

;; -----
;;  FIND NEAREST NEIGHBOR
;;  -----

to find-nearest-neighbor
  set nearest-neighbor min-one-of schoolmates [ distance myself ]
end

;; -----
;;  SEPARATION RULE (AVOID COLLISION)
;;  -----

to separate
  ;; Turn away from nearest neighbor
  let dx (dx-of nearest-neighbor)
  let dy (dy-of nearest-neighbor)
  rt (random 20 + 160)
  fd 0.1
end

;; -----
;;  COHESION RULE (TURN TOWARD GROUP)
;;  -----

```



```
to cohere
  if any? schoolmates [
    face mean-heading-of schoolmates
  ]
end

;; -----
;; ALIGNMENT RULE (MATCH GROUP HEADING)
;; -----

to align
  if any? schoolmates [
    set heading (heading + (mean [heading] of schoolmates - heading) * 0.1)
  ]
end

;; -----
;; SPEED ADJUSTMENT (MAINTAIN COHESION)
;; -----

to adjust-speed
  ifelse max [speed] of schoolmates > speed + 0.1
  [
    ;; Accelerate to keep up with group
    set speed speed + 0.1
  ]
  [
    ;; Slow down toward minimum speed
    if speed > min-speed [
      set speed speed - 0.2
    ]
  ]
end
```


Chapter 13

Selective Tidal Stream Transport

13.1 Overview

Selective Tidal Stream Transport (STST) is a behavioral strategy that enables agents to conserve energy by alternating between active swimming and passive drift under strong hydrodynamic forcing. Within this submodel, STST is triggered when the along-channel velocity of a patch exceeds an agent's effective swimming speed and that speed falls below a species-specific minimum threshold. When engaged, agents align with the tidal current and are passively transported upstream or downstream depending on flow direction, reducing the metabolic cost of movement by substituting active swimming with drift.

This behavior is well documented in anadromous and estuarine fishes, where individuals exploit tidal currents to facilitate migration while minimizing energy expenditure (??). Movement during STST is governed by local flow velocity, allowing migration pathways, displacement, and energy savings to emerge dynamically from the interaction between individual swimming capacity and the tidal flow field. STST persists for a limited duration or until swimming ability improves, after which agents resume directional migration.

13.2 Purpose

To simulate a passive energy-conserving behavior in migratory fish that allows them to use tidal currents to move when swimming capacity is insufficient to overcome flow velocities.

13.3 Entities, State Variables, and Scales

13.3.1 Spatial and Temporal Scales

Spatial Unit: Patch (3 m x 3 m resolution)

Temporal Unit: 5 minute time steps (tick)

13.3.2 Global Variables

Variable Name	Definition
max-seaward-velocity	Highest magnitude seaward-directed hydrodynamic velocity across the spatial domain.
max-landward-velocity	Highest magnitude landward-directed hydrodynamic velocity across the spatial domain.

13.3.3 Patch Variables

Variable Name	Definition
Velocity V_{patch}	The along-channel velocity of a given patch, derived from hydrodynamic model inputs, where positive values are in the landward direction and negative values are in the seaward direction.
tidal-transport-in-patch	Count of agents exhibiting tidal stream transport within a patch (for habitat quality analysis).

13.3.4 Agent Variables

Variable Name	Definition
energy E_{agent}	Total energy available to the agent.
swimming energy E_{swim}	Energy expenditure from movement per time step.
base swim energy $swim_{base}$	Baseline energy cost of movement.
swimming difficulty D_f	Velocity-based proxy representing hydrodynamic resistance.
in-STST? $STST?$	Boolean value indicating if the agent is actively in STST.

swimming speed V_{agent}	The effective swimming speed of the agent.
minimum threshold speed $Speed_{min}$	The minimum speed at which an agent will move.

13.4 Process Overview and Scheduling

1. Compare swimming speed (A_{speed}) with flow speed (V_{patch}).
2. If $|V_{patch}| > A_{speed}$ and $A_{speed} \leq Speed_{min}$, enter STST.
3. In STST: align with current, update position via drift, apply reduced energy cost.
4. If $A_{speed} > Speed_{min}$, exit STST and resume active swimming.

13.5 Design Concepts

Basic Principles: Selective tidal stream transport (STST) is a documented behavioral strategy in anadromous fishes that exploits tidal currents to conserve energy during migration (e.g., ?; ?)

Emergence: Passive drift behavior and resulting migration paths emerge from agent-flow interactions and individual swimming limitations.

Adaptation: Agents adapt their mode of movement based on their swimming ability relative to environmental flow, dynamically choosing energy-efficient strategies.

Objectives: Agents seek to minimizing energy loss in strong flows.

Sensing: Agents sense their own A_{speed} and the V_{patch} to determine whether passive drift is needed.

Observation: Records STST patch events, energy expenditure (E_{agent}), and displacement are logged to analyze behavior across flow regimes.

13.6 Initialization

Variable	Initialized Value	Justification
V_{agent}	Based on size, energy, and difficulty factor	Reflects agent's swimming capability based on metabolic limits.

Variable	Initialized Value	Justification
$swim_{min}$	Species-specific parameter	Represents the minimum sustained swimming velocity of the agent.
$swim_{max}$	Species-specific parameter	Represents the maximum sustained swimming velocity of the agent.
E_{agent}	100	Assumes full energy at the start of simulation or at spawning.
$swim_{base}$	$0.02 \cdot \frac{M_{agent}^k}{M_{max}^k}$	Scales locomotion cost nonlinearly with size; can be calibrated.

13.7 Submodels

13.7.1 Trigger Conditions for STST

Agents compare their swimming ability to the flow conditions. If local flow exceeds their capability and their effort is below a defined threshold, they enter STST:

$$|V_{patch}| > A_{speed} \quad \text{and} \quad A_{speed} \leq Speed_{min} \quad (\text{C.94})$$

Where:

- V_{patch} is the depth-Averaged along-channel water velocity at the patch (m s^{-1}).
- A_{speed} is the actual swimming speed used for movement (m s^{-1}).
- $Speed_{min}$ is the minimum sustainable swimming speed (m s^{-1}).

While in STST:

Heading aligns with the current (drift vector) & swimming speed is set to:

$$A_{speed} = |V_{patch}| \quad (\text{C.95})$$

Where:

- V_{patch} is the depth-Averaged along-channel water velocity at the patch (m s^{-1}).
- A_{speed} is the actual swimming speed used for movement (m s^{-1}).

Energy is set as:

$$E_{swim} = swim_{base} \quad (C.96)$$

Where:

- E_{swim} is the energy cost of swimming for the tick (energy units).
- $swim_{base}$ is the baseline energetic cost of swimming (energy units).

13.7.2 Behavior During STST

While in STST, agents align with the current (either landward or seaward) and are passively transported:

$$\vec{Y}_{t+1} = \vec{Y}_t + |V_{patch}| \cdot \hat{u} \quad (C.97)$$

Where:

- $\vec{Y}_t$ is the Agent spatial position vector.
- V_{patch} is the depth-Averaged along-channel water velocity at the patch (m s⁻¹).
- $\hat{u}_{patch}$ is the unit direction vector of patch flow, defined as $\hat{u}_{patch} = V_{patch}/|V_{patch}|$ when $|V_{patch}| > 0$.

Swimming speed is overwritten:

$$A_{speed} = |V_{patch}| \quad (C.98)$$

Where:

- A_{speed} is the actual swimming speed used for movement (m s⁻¹).
- V_{patch} is the depth-Averaged along-channel water velocity at the patch (m s⁻¹).

Energy cost is minimized:

$$E_{swim} = swim_{base} \quad (C.99)$$

Where:

- E_{swim} is the energy cost of swimming for the tick (energy units).
- $swim_{base}$ is the baseline energetic cost of swimming (energy units).

13.7.3 Stop Conditions for STST

Agents exit STST when they regain sufficient swimming capacity to exceed threshold:

$$A_{speed} > speed_{min} \quad (C.100)$$

Where:

- A_{speed} is the actual swimming speed used for movement (m s^{-1}).
- $speed_{min}$ is the minimum sustainable swimming speed (m s^{-1}).

This triggers a return to active migratory movement and deactivates $STST_7$.

13.8 Netlogo Implementation

```
;; -----
;; GLOBALS
;; -----
globals [
  max-seaward-velocity
  max-landward-velocity

  landward-migration?
  seaward-migration?

  selective-tidal-transport?
  STST-start-tick
]

;; -----
;; PATCH VARIABLES
;; -----
patches-own [
  velocity
  depth
  patch-terrain

  tidal-transport-in-patch
]

;; -----
;; TURTLE VARIABLES
;; -----
turtles-own [
  speed
```



```

min-speed
swim-efficiency
swim-base
difficulty-factor
energy
E-swim

selective-tidal-transport?
STST-start-tick
]

;; -----
;; STST - LANDWARD
;; -----
to selective-tidal-stream-transport-landward
  let V-patch [velocity] of patch-here
  let S-swim (speed / 300) ;; 5 minute speed

  ;; Continue drifting (1 tick cooldown)
  if selective-tidal-transport? and (ticks - STST-start-tick < 1) [
    let drift-distance abs V-patch * (1 - swim-efficiency)
    let drift-target one-of neighbors with [patch-terrain = "water"]
    if drift-target != nobody [ safe-move drift-target drift-distance ]
    set tidal-transport-in-patch tidal-transport-in-patch + 1
    stop
  ]

  ;; Enter STST
  if abs V-patch > S-swim and S-swim <= min-speed [
    set selective-tidal-transport? true
    set STST-start-tick ticks
    let drift-distance abs V-patch * (1 - swim-efficiency)
    let drift-target one-of neighbors with [patch-terrain = "water"]
    if drift-target != nobody [ safe-move drift-target drift-distance ]
    set tidal-transport-in-patch tidal-transport-in-patch + 1
    stop
  ]

  ;; Exit STST → resume active migration
  if S-swim > min-speed or (ticks - STST-start-tick >= 1) [
    set selective-tidal-transport? false
    calculate-swim-energy
    set energy max list 0 (energy - E-swim)
    migrate-landward
  ]
]

```

```

end

;; -----
;; STST - SEAWARD
;; -----
to selective-tidal-stream-transport-seaward
  let V-patch [velocity] of patch-here
  let S-swim (speed / 300)

  ;; Continue STST drift
  if selective-tidal-transport? and (ticks - STST-start-tick < 1) [
    let drift-distance abs V-patch * (1 - swim-efficiency)
    let drift-target one-of neighbors with [patch-terrain = "water"]
    if drift-target != nobody [ safe-move drift-target drift-distance ]
    set tidal-transport-in-patch tidal-transport-in-patch + 1
    stop
  ]

  ;; Enter STST
  if abs V-patch > S-swim and S-swim <= min-speed [
    set selective-tidal-transport? true
    set STST-start-tick ticks
    let drift-distance abs V-patch * (1 - swim-efficiency)
    let drift-target one-of neighbors with [patch-terrain = "water"]
    if drift-target != nobody [ safe-move drift-target drift-distance ]
    set tidal-transport-in-patch tidal-transport-in-patch + 1
    stop
  ]

  ;; Exit STST → resume seaward migration
  if S-swim > min-speed or (ticks - STST-start-tick >= 1) [
    set selective-tidal-transport? false
    calculate-swim-energy
    set energy max list 0 (energy - E-swim)
    migrate-seaward
  ]
end

;; -----
;; SAFE MOVEMENT HANDLER
;; -----
to safe-move [target-patch move-speed]
  face target-patch
  let next-patch patch-ahead move-speed

```

```
ifelse [patch-terrain] of next-patch = "water" [  
  fd move-speed  
] [  
  let nearby-water one-of patches in-radius 1 with [patch-terrain = "water"]  
  if nearby-water != nobody [ move-to nearby-water ]  
]  
end
```


Chapter 14

Predation and Fleeing Behavior

14.1 Overview

This module simulates predator–prey interactions between striped bass (predators) and alewife (prey) during migratory and staging periods, representing encounters as localized behavioral interactions embedded within movement, schooling, and energetic state. Predators visually detect nearby prey based on proximity, visual range, gape limitation, and stomach capacity, initiate pursuit when energetically feasible, and consume prey within capture range. Prey respond through directional fleeing and short evasive movements, with escape responses propagating across nearby individuals through a patch-level alarm mechanism. The model tracks energetic costs of pursuit and escape, prey consumption events, and resulting spatial patterns of predation risk.

The interaction structure is adapted from the NetLogo schooling and predation framework of (?), extended here by linking chase, escape, and capture outcomes to hydrodynamic movement, energetic cost, and contaminant transfer. Predator–prey interactions are inherently spatial and emerge from movement and local overlap among individuals, where encounter rates are governed by proximity and relative motion through the environment (?). These interactions follow established predator–prey theory in which consumption arises from encounter processes and functional responses linking prey availability to predator intake (?), allowing predation risk, prey displacement, predator foraging success, and contaminant uptake to emerge dynamically rather than being imposed as fixed mortality or feeding rates.

14.2 Purpose and Patterns

This submodel captures behaviorally realistic predator–prey dynamics relevant to estuarine migration and staging.

- Prey initiate fleeing when striped bass are within a defined visual neighborhood and prey have sufficient energy to respond.
- Predators initiate pursuit when prey are visible and the predator is not constrained by energetic state or stomach capacity.
- Fleeing behavior changes prey movement mode and overrides normal migration movement.
- Predation events occur when predators and prey co locate within a capture neighborhood and prey meet gape constraints.
- Predation risk hotspots emerge from the overlap of predator movement, prey distribution, and hydrodynamic movement constraints.

14.3 Entities, State Variables, and Scales

14.3.1 Spatial and Temporal Scales

Spatial Unit: Patch (3 m x 3 m resolution)

Temporal Unit: 5 minute time steps (tick)

14.3.2 Patch Variables

Variable Name	Definition
<code>patch-terrain</code>	Patch classification used to restrict movement to water patches.
<code>prey-eaten-in-patch</code>	Count of prey consumed in a patch.
<code>visits-by-alewife</code>	Cumulative count of prey patch visits, used for exposure and space use summaries.
<code>ticks-spent-alewife</code>	Cumulative time spent by prey on a patch.

14.3.3 Prey: Agent Variables

Variable Name	Definition
<code>fleeing?</code>	Boolean indicator of whether the prey is actively fleeing.

Variable Name	Definition
energy	Energetic state used to determine whether prey can flee and to regulate behavior switching.
speed	Current movement speed used in the swim procedure.
max-speed	Maximum sustained swim speed.
vision-distance	Detection or interaction distance used in predator encounter checks.
planned-path	Stored movement path used during fleeing movement.
trail	List of visited patches retained for optional visualization.
previous-patch, previous-x, previous-y	Tracking variables supporting movement diagnostics and behavior switching.

14.3.4 Predator: Agent Variables

Variable Name	Definition
vision-distance	Radius within which prey are considered visible to the predator.
prey-in-vision	Agentset of visible prey within the predator's vision radius.
hunting?	Boolean indicator of whether the predator is actively pursuing prey.
bursting?	Boolean indicator of whether the predator is in burst speed mode.
speed	Predator movement speed, which can increase during bursting.
max-speed	Maximum predator speed used during chase.
stomach-contents	Current stomach contents used to limit feeding and trigger foraging and lipid use when empty.
stomach-capacity	Maximum stomach capacity.
gape-size	Predator gape size used as a constraint on edible prey.
daytime-prey-eaten	Daily counter of prey eaten.
total-prey-eaten	Cumulative prey eaten (tracked via counters).
prey-eaten-in-patch	Patch level counter incremented during predation.

Variable Name	Definition
E-swim	Swimming energy cost computed during movement.
difficulty-factor	Hydrodynamic movement difficulty used to scale energetic cost of movement.
swim-efficiency	Efficiency factor used to scale energetic cost of movement.

14.4 Process Overview and Scheduling

Prey Behavior (per agent)

1. During **swim**, if **fleeing?** is false, prey execute normal migration movement and pay swimming energetic costs.
2. If **fleeing?** is true, prey execute **flee-strippedbass**, which overrides normal migration movement and moves prey through a short random evasive path over water patches.
3. Prey stop fleeing when no striped bass are detected within the configured encounter range or when prey energy is depleted.

Predator Behavior (per agent)

1. During active movement periods, predators define $\mathcal{P}_{vision}$ as prey within A_{vision} .
2. If stomach contents are empty, predators enter foraging mode and may activate lipid loss depending on the broader energy logic.
3. If any prey are in vision, predators set **hunting?** to true, chase prey, and attempt capture.
4. Predators adjust speed during pursuit, trigger prey fleeing behaviors, and consume prey when capture criteria are met.
5. Predation updates predator stomach contents, prey counters, and patch level consumption counters.

14.5 Design Concepts

Basic Principles: Predator–prey interactions are adapted from the (?) Net-Logo schooling and predation framework. The core structure of local visual detection, directional pursuit, prey fleeing, and stochastic capture is retained from that implementation. In P-MEM, these interaction rules are extended by coupling chase and escape behavior to energetic constraints, gape limitation, stomach capacity, and contaminant transfer. Thus, the underlying encounter

mechanics broadly follow ?, while cost-routing, physiological scaling and contaminant dynamics represent model extensions.

Emergence: Spatial predation patterns emerge from coupled movement processes, including prey migration routes, predator search behavior, and the temporary displacement of prey caused by fleeing responses.

Adaptation: Prey adaptively switch from migration movement to fleeing movement when predators are nearby. Predators adaptively switch between hunting and wandering depending on prey availability and stomach state.

Objectives: Predators maximize energetic intake by capturing prey when available while managing energetic costs of pursuit. Prey minimize predation risk by fleeing when threatened, balanced against energetic limitations.

Sensing: Predators and prey sense one another through neighborhood or cone based proximity checks using A_{vision} . Prey fleeing is triggered by local predator presence.

Stochasticity: Stochasticity arises through random turning and patch selection during fleeing movement and through prey selection during capture attempts.

Collectives: Collective dynamics are represented indirectly through the simultaneous triggering of fleeing across multiple prey within the predator's encounter neighborhood.

Observation: Key outputs include prey consumption rates, prey and predator energetic trajectories, and time periods of predation, and the spatial distribution of predation events ($P_{eaten,patch}$).

14.6 Initialization

Variable	Initialized Value	Justification
energy	100	Prey begin fully energized to allow immediate fleeing or normal movement
fleeing?	false	Prey are not actively escaping at the start
prey-alarmed?	false	Alarm cue inactive at start of simulation
daytime-prey-eaten	0	daily counter of prey eaten starts at zero
time-since-full	0	predator begins simulation hungry
gape-limit	function of size	Large predators have wider gape limits

Variable	Initialized Value	Justification
handling-effort	calculated per encounter	Depends on prey traits and local patch conditions
reaction-time	size-, age-, and SPM-based	Prey and predator values dynamically computed each tick
flee-ability	based on age	Represents individual escape competence across age classes
predation-ability	based on age	Mid-age predators have fastest and most accurate prey responses
$swim_{max}$	$1.5 - 3 \frac{bodylengths}{sec}$	Typical value for sustained swimming speed in small pelagic fish (refer to Videler, 1993).

14.7 Submodels

14.7.1 Prey Fleeing Trigger and Directional Response

Prey enter a fleeing state when a striped bass is detected within a cone-based encounter neighborhood and prey energy is positive:

$$\exists predator \in \mathcal{C}_{prey}(r, \theta) \wedge E_{agent} > 0 \Rightarrow fleeing? = \text{true} \quad (\text{C.101})$$

Where:

- $\mathcal{C}_{prey}(r, \theta)$ is the prey encounter cone with radius $r = A_{vision}$ (patch units) and cone angle θ (implemented as `in-cone A_{vision} 180`) (degrees).
- E_{agent} is the total internal energy available to the agent (patch units).

Prey exit fleeing when no striped bass are detected within the encounter neighborhood or when energy is depleted:

$$\neg \exists predator \in \mathcal{C}_{prey}(r, \theta) \vee E_{agent} \leq 0 \Rightarrow fleeing? = \text{false} \quad (\text{C.102})$$

Where:

- $\mathcal{C}_{prey}(r, \theta)$ is the prey encounter cone with radius $r = A_{vision}$ (patch units) and cone angle θ (implemented as `in-cone A_{vision} 180`) (degrees).

- E_{agent} is the total internal energy available to the agent (energy units).

Directional escape is implemented by assigning heading based on predator relative position, corresponding to the **scare-prey** control flow:

$$\psi_{t+1} = \begin{cases} \psi_t + 90^\circ & (\text{scare-right}) \\ \psi_t - 90^\circ & (\text{scare-left}) \\ \psi_t + 180^\circ & (\text{scare-down}) \\ \psi_t + 0^\circ & (\text{scare-up}) \end{cases} \quad (\text{C.103})$$

Where:

- $\mathcal{C}_{prey}(r, \theta)$ is the prey encounter cone with radius $r = A_{vision}$ (patch units) and cone angle θ (implemented as **in-cone** `A_{vision}` 180) (degrees).
- ψ is the agent heading (degrees).

14.7.2 Prey Fleeing Movement and Energetic Cost

When fleeing, prey move at maximum speed:

$$A_{speed} = speed_{max} \quad (\text{C.104})$$

Where:

- A_{speed} is the actual swimming speed used for movement (m s^{-1}).
- $speed_{max}$ is the maximum achievable swimming speed (m s^{-1}).

Travel distance per tick is computed from current speed (with the model's internal conversion):

$$D_{travel} = \frac{A_{speed}}{3} \quad (\text{C.105})$$

Where:

- D_{travel} is the distance traveled in the current tick (patch units), where 1 patch = 3 m.
- A_{speed} is the actual swimming speed used for movement (m tick^{-1}).

A small random turn is applied:

$$\psi_{t+1} = \psi_t + \varepsilon, \quad \varepsilon \sim U(-15^\circ, 15^\circ) \quad (\text{C.106})$$

Where:

- ψ is the agent heading (degrees).
- ε is a random turning increment (degrees).

A target patch is selected from reachable water patches within distance D_{travel} :

$$p^* \sim \text{Uniform}(\{p \in \mathcal{N}(d_t) : \text{patch-terrain}(p) = \text{"water"}\}) \quad (\text{C.107})$$

Where:

- p^* is the selected target patch (patch location).
- p is the patch occupied by the prey agent (patch location).
- $\mathcal{N}(D_{travel})$ is the set of patches within radius D_{travel} .
- $\text{patch-terrain}(p)$ returns the terrain type of patch p .
- $\text{Uniform}(\cdot)$ indicates uniform random selection.

Prey traverse a path from the current patch p to p^* (implementation uses `build-path-to-prey`) and move along up to

$$n_t = \lceil D_{trave} \rceil \quad (\text{C.108})$$

Where:

- n_t is the number of patch steps taken within the tick (patches).
- $\lceil \cdot \rceil$ denotes the ceiling operator.

patch steps within the tick. Patch visitation and time spent are accumulated:

$$\text{visits-by-alewife}(p) \leftarrow \text{visits-by-alewife}(p) + 1 \quad (\text{C.109})$$

$$\text{ticks-spent-alewife}(p) \leftarrow \text{ticks-spent-alewife}(p) + 1 \quad (\text{C.110})$$

Where:

- p is the patch occupied by the prey agent (patch location).
- $\text{visits-by-alewife}(p)$ records cumulative visitation (visits).
- $\text{ticks-spent-alewife}(p)$ records cumulative residence time (ticks).

Swimming energetic cost is applied through the shared swimming energy routine. First compute an energy multiplier from hydrodynamic difficulty and swimming efficiency:

$$M_{swim} = D_f \cdot \left(\frac{1}{\text{swim}_{Eff}} \right) \quad (\text{C.111})$$

Where:

- D_f is mapped swimming difficulty index (dimensionless).

- $swim_{Eff}$ is the efficiency multiplier applied to swimming energetics (%).
- M_{swim} is the swimming energy multiplier (dimensionless).

Then compute swimming energy cost (exponential scaling):

$$E_{swim} = swim_{base} \cdot m_{swim}^{\beta} \quad (C.112)$$

Where:

- E_{swim} is the energy cost of swimming for the tick (energy units).
- $swim_{base}$ is the baseline energetic cost of swimming ((energy units) patch⁻¹).
- M_{swim} is the swimming energy multiplier (dimensionless).
- β is the exponent controlling swimming cost scaling (dimensionless).

with $\beta = 0.75$. Energy is updated and constrained to remain non-negative:

$$energy_{t+1} = \max(0, energy_t - E_{swim,t}) \quad (C.113)$$

Where:

- D_f is the mapped swimming difficulty index (dimensionless).
- $swim_{Eff}$ is the efficiency multiplier applied to swimming energetics (%).
- M_{swim} is the swimming energy multiplier (dimensionless).
- $swim_{base}$ is the baseline energetic cost of swimming ((energy units) patch⁻¹).
- β is the exponent controlling swimming cost scaling (dimensionless).
- E_{agent} is the total internal energy available to the agent (energy units).

14.7.3 Predator Prey Detection and Chase

Predators define visible prey as prey within a radius equal to A_{vision} :

$$\mathcal{P}_{vision} = \{i \in \text{prey} : \|x_i - x_{pred}\| \leq A_{vision} \wedge patchtype(i) \neq \text{"sea"}\} \quad (C.114)$$

Where:

- $\mathcal{P}_{vision}$ is the set of prey visible to the predator.
- i indexes individual prey agents.
- x_i is the spatial position vector of prey i (patch coordinates).
- x_{pred} is the spatial position vector of the predator (patch coordinates).
- $\|\cdot\|$ denotes Euclidean distance (patch units).
- A_{vision} is the predator's visual detection radius (patch units).
- $patchtype(i)$ returns the terrain type occupied by prey i .

Predators enter hunting mode when any prey are visible:

$$|\mathcal{P}_{vision}| > 0 \Rightarrow hunting? = \text{true} \quad (\text{C.115})$$

Where:

- $\mathcal{P}_{vision}$ is the set of prey visible to the predator.
- $hunting?$ is the indicator that predator is hunting.

If hunting, the predator targets the nearest visible prey:

$$i^* = \arg \min_{i \in \mathcal{P}_{vision}} \|x_i - x_{pred}\| \quad (\text{C.116})$$

Where:

- i^* is the selected target prey.
- $\arg \min$ returns the prey minimizing distance to the predator (patch units).
- x_i is the spatial position vector of prey i (patch coordinates).
- x_{pred} is the spatial position vector of the predator (patch coordinates).

Predator movement toward prey is computed using a step length limited by travel distance per tick and prey distance:

$$D_{travel} = \frac{A_{speed}}{3} \quad (\text{C.117})$$

Where:

- D_{travel} is the distance traveled in the current tick (patch units), where 1 patch = 3 m.
- A_{speed} is the actual swimming speed used for movement (m tick⁻¹).

$$step_t = \min(D_{travel}, \|x_{i^*} - x_{pred}\|) \quad (\text{C.118})$$

Where:

- $step_t$ is the realized movement distance toward the selected target during the current tick.
- D_{travel} is the distance traveled in the current tick (patch units), where 1 patch = 3 m.
- x_i is the spatial position vector of prey i (patch coordinates).
- x_{pred} is the spatial position vector of the predator (patch coordinates).

$$x_{pred+1} = x_{pred} + step_t \cdot \hat{u}(x_{i^*} - x_{pred}) \quad (\text{C.119})$$

Where:

- x_{pred} is the spatial position vector of the predator (patch coordinates).
- $step_t$ is the realized movement distance toward the selected target during the current tick (patch units).
- $\hat{u}$ is the unit direction vector toward the target prey (dimensionless).
- x_{i^*} is the spatial position vector of prey i (patch coordinates).

Swimming energetic cost during chase is applied using the same shared routine:

$$E_{swim,t} = swim_{base} \cdot \left(D_f \cdot \frac{1}{swim_{Eff}} \right)^\beta \quad (C.120)$$

Where

- E_{swim} is the energy cost of swimming for the tick (energy units).
- D_f is the mapped swimming difficulty index (dimensionless).
- $swim_{Eff}$ is the efficiency multiplier applied to swimming energetics (%).
- $swim_{base}$ is the baseline energetic cost of swimming ((energy units) patch⁻¹).
- β is the exponent controlling swimming cost scaling (dimensionless).

$$E_{agent+1} = \max(0, E_{agent} - E_{swim,t}) \quad (C.121)$$

Where:

- $\mathcal{P}_{vision}$ is the set of visible prey.
- x_{i^*} is the spatial position vector of prey i (patch coordinates).
- $\hat{u}$ is the unit direction vector toward prey (patch coordinates).
- $step_t$ is the realized movement distance toward the selected target during the current tick (patch units).
- E_{agent} is the total internal energy available to the agent (energy units).
- E_{swim} is the energy cost of swimming for the tick (energy units).

14.7.4 Predator Burst Speed Adjustment

Predators define visible prey as prey within a radius equal to A_{vision} :

$$\mathcal{P}_{vision} = \{i \in \text{prey} : \|x_{i^*} - x_{pred}\| \leq A_{vision} \wedge patchtype(i) \neq \text{"sea"}\} \quad (C.122)$$

Where:

- A_{vision} is the visual detection radius of agent (patch units).
- $\mathcal{P}_{vision}$ is the set of visible prey.
- x_{i^*} is the spatial position vector of prey i (patch coordinates).
- x_{pred} is the spatial position vector of the predator (patch coordinates).
- $patchtype$ is the location classifier of the patch.

Predators enter hunting mode when any prey are visible:

$$|\mathcal{P}_{vision}| > 0 \Rightarrow hunting? = \text{true} \quad (\text{C.123})$$

Where:

- $\mathcal{P}_{vision}$ is the set of visible prey.
- $hunting?$ is the indicator that predator is hunting.

If hunting, the predator targets the nearest visible prey:

$$i^* = \arg \min_{i \in \mathcal{P}_{vision}} \|x_{i^*} - x_{pred}\| \quad (\text{C.124})$$

Where:

- i^* is the selected target prey.
- $\mathcal{P}_{vision}$ is the set of visible prey.
- x_{i^*} is the spatial position vector of prey i (patch coordinates).
- x_{pred} is the spatial position vector of the predator (patch coordinates).

Predator movement toward prey is computed using a step length limited by travel distance per tick and prey distance:

$$D_{travel} = \frac{A_{speed}}{3} \quad (\text{C.125})$$

Where:

- D_{travel} is the distance traveled in the current tick (patch units), where 1 patch = 3 m.
- A_{speed} is the actual swimming speed used for movement (m tick⁻¹).

$$step_t = \min(D_{travel}, \|x_{i^*} - x_{pred}\|) \quad (\text{C.126})$$

Where:

- $step_t$ is the realized movement distance toward the selected target during the current tick.
- D_{travel} is the distance traveled in the current tick (patch units), where 1 patch = 3 m.
- x_{i^*} is the spatial position vector of prey i (patch coordinates).
- x_{pred} is the spatial position vector of the predator (patch coordinates).

$$x_{pred+1} = x_{pred} + step_t \cdot \hat{u}(x_{i^*} - x_{pred}) \quad (\text{C.127})$$

Where:

- x_{pred} is the spatial position vector of the predator (patch coordinates).
- $step_t$ is the realized movement distance toward the selected target during the current tick (patch units).
- $\hat{u}$ is the unit direction vector toward prey.
- x_{i^*} is the spatial position vector of prey i (patch coordinates).

Swimming energetic cost during chase is applied using the same shared routine:

$$E_{swim} = swim_{base} \cdot \left(D_f \cdot \frac{1}{swim_{Eff}} \right)^\beta \quad (C.128)$$

Where:

- E_{swim} is the energy cost of swimming for the tick (energy units).
- D_f is the mapped swimming difficulty index (dimensionless).
- $swim_{Eff}$ is the efficiency multiplier applied to swimming energetics (%).
- $swim_{base}$ is the baseline energetic cost of swimming ((energy units) patch⁻¹).
- β is the exponent controlling swimming cost scaling.

$$E_{agent+1} = \max(0, E_{agent} - E_{swim}) \quad (C.129)$$

Where:

- E_{agent} is the total internal energy available to the agent (energy units).
- E_{gain} is the energy gained from digested biomass (energy units).

14.7.5 Predation and Consumption

Predators attempt capture only if stomach contents are below Predators can successfully capture only if stomach contents are below capacity:

$$M_{stomach} < stomach_{max} \quad (C.130)$$

Where:

- $M_{stomach}$ is the biomass currently stored in the stomach (g).
- $stomach_{max}$ is the maximum stomach capacity (g).

Candidate prey include prey on the same patch and neighboring patches (local capture neighborhood):

$$\mathcal{P}_{near} = \{i \in \text{prey} : i \in \{here\} \cup \{neighbors\}\} \quad (C.131)$$

Where:

- $\mathcal{P}_{near}$ is a set of prey individuals located on the predator's current patch.

- i is the index representing an individual prey agent.

Edible prey satisfy the gape constraint:

$$\mathcal{P}_{edible} = \{i \in \mathcal{P}_{near} : body-depth_i \leq gape-size_{pred}\} \quad (C.132)$$

Where:

- $\mathcal{P}_{edible}$ is a subset of nearby prey that satisfy the predator's gape limitation and are physically consumable.
- $\mathcal{P}_{near}$ is a set of prey individuals located on the predator's current patch.
- $body-depth_i$ is the body depth of the selected prey individual (mm).
- $gape-size_{pred}$ is the maximum ingestible prey depth for predator (mm).
- i is the index representing an individual prey agent.

If any edible prey exist, one prey is selected:

$$i^* \sim \text{Uniform}(\mathcal{P}_{edible}) \quad (C.133)$$

Where:

- i^* is the selected target prey.
- $\mathcal{P}_{edible}$ is a subset of nearby prey that satisfy the predator's gape limitation and are physically consumable.

Digestible meal mass is assumed to be 75% of prey wet weight:

$$M_{meal} = 0.75 \cdot A_{mass,i^*} \quad (C.134)$$

Where:

- M_{meal} is the digestible biomass of captured prey (g).
- A_{mass,i^*} is the agent body mass of the selected target prey (g).

Available stomach space is:

$$S_{stomach} = stomach_{max} - M_{stomach} \quad (C.135)$$

Where:

- $S_{stomach}$ is the available stomach capacity (g).
- $stomach_{max}$ is the maximum stomach capacity (g).
- $M_{stomach}$ is the biomass currently stored in the stomach (g).

Intake is limited by available space:

$$M_{intake} = \min(M_{meal}, S_{stomach}) \quad (C.136)$$

Where:

- M_{intake} is the biomass actually ingested by the agent (g).
- M_{meal} is the digestible biomass of captured prey (g).
- $S_{stomach}$ is the available stomach capacity (g).

Stomach contents are updated:

$$M_{stomach+1} = M_{stomach} + M_{intake} \quad (C.137)$$

Where:

- $S_{stomach}$ is the available stomach capacity (g).
- M_{intake} is the biomass actually ingested by the agent (g).

Prey-derived contaminant loads are added to stomach contaminant pools:

$$stomach\text{-}contents\text{-}MeHg_{t+1} = stomach\text{-}contents\text{-}MeHg_t + MeHg_{i^*} \quad (C.138)$$

Where:

- $S_{stomach}$ is the available stomach capacity (g).
- $MeHg_{i^*}$ methylmercury content of target prey (ng).

Consumption counters are incremented:

$$\mathcal{P}_{eaten+1} = \mathcal{P}_{eaten} + 1 \quad (C.139)$$

Where:

- $\mathcal{P}_{eaten}$ is the total prey consumed by a predator.

$$\mathcal{P}_{eaten,patch+1} = \mathcal{P}_{eaten,patch} + 1 \quad (C.140)$$

Where:

- $\mathcal{P}_{eaten,patch}$ is the amount of prey consumed on a given patch.

Selected prey is removed from the simulation:

$$alive(i^*) \leftarrow 0 \quad (C.141)$$

Where:

- $alive(i^*)$ is the Boolean indicating whether selected prey is alive (0,1).

14.7.6 Interaction With Migration Movement and Swimming Energy

Movement and swimming energetic cost are handled within the shared movement controller `swim`. If prey are not fleeing, migration movement is executed through STST procedures and swimming energy is charged. If prey are fleeing, fleeing movement is executed and swimming energy is charged. Therefore, no additional movement energy term is added within predation or fleeing procedures:

$$E_{move} = E_{swim} \quad \text{applied once per tick via } \textit{calculate-swim-energy} \quad (\text{C.142})$$

Where:

- E_{move} is the energy cost to move toward prey (energy units).
- E_{swim} is the energy cost of swimming for the tick (energy units).

Chapter 15

Model Building Tutorial: Staging & Schooling

15.0.1 Module Integration

- Brief explanation of how functions (e.g., staging, schooling) interact.
- Clarify temporal structure (e.g., tick-based sequence) and spatial scale.
- Describe coupling logic

15.0.2 Function Dependencies

- What variables are required as inputs for each function?
- What functions must be called before/after? (e.g., must calculate stress before checking staging triggers)
- Dependency table showing variable flow between submodels.

Function	Required Inputs	Output Variables	Dependent On
staging	E_agent, I_stress	E_agent, patch records	osmoregulation
schooling			

15.1 Implementation in NetLogo:

Programming details

example code

15.2 Implementation in R:

Programming details

example code

Chapter 16

Model Building Tutorial: Landward Migration & Selective Tidal Stream Transport

16.0.1 Module Integration

- Brief explanation of how functions (e.g., staging, schooling) interact.
- Clarify temporal structure (e.g., tick-based sequence) and spatial scale.
- Describe coupling logic

16.0.2 Function Dependencies

- What variables are required as inputs for each function?
- What functions must be called before/after? (e.g., must calculate stress before checking staging triggers)
- Dependency table showing variable flow between submodels.

Function	Required Inputs	Output Variables	Dependent On
osmoregulation	S_patch, S_agent, C	I_stress, C_new, E_osmo	environmental-sensing
staging	E_agent, I_stress	E_agent, patch records	osmoregulation
migration	E_agent, V_patch	Y_t, E_agent	staging, STST

Function	Required Inputs	Output Variables	Dependent On
STST	V_patch, V_agent	Y_t+1, E_agent	migration logic

16.1 Implementation in NetLogo:

Programming details
example code

16.2 Implementation in R:

Programming details
example code

Chapter 17

Complex Model Building Tutorial: One-Way Migration

17.0.1 Module Integration

- Brief explanation of how functions (e.g., staging, schooling) interact.
- Clarify temporal structure (e.g., tick-based sequence) and spatial scale.
- Describe coupling logic

17.0.2 Function Dependencies

- What variables are required as inputs for each function?
- What functions must be called before/after? (e.g., must calculate stress before checking staging triggers)
- Dependency table showing variable flow between submodels.

17.1 Implementation in NetLogo:

Programming details

example code

17.2 Implementation in R:

Programming details

example code

Chapter 18

Modeling Toolkit

18.1 Learning Resources

- **NetLogo Library:** Brief note on structure (e.g., model categories), how you used it (e.g., behavior ideas, calibration).
- **NetLogo User Manual:** Link to official documentation.
- **NetLogo Modeling Commons:** Peer-contributed models, code sharing, and idea sourcing.
- **NetLogo Forum:** Where to get help or search issues.
- **Book References:**
 - Railsback & Grimm (2019) **Agent-Based and Individual-Based Modeling**
 - Grimm & Railsback (2005) **Individual-Based Modeling and Ecology**
 - Add others specific to modeling.

18.2 Best Practices

- **ODD Protocol (Overview, Design concepts, Details):** Follow for transparent and structured documentation of agent-based models. Ensures clarity across entities, processes, and assumptions.
- **Modular Design:** Structure behavioral functions into separate procedures (e.g., swimming, osmoregulation, staging) to support testing, integration, and reuse.
- **Version Control:** Use GitHub or other tools to track changes and link code to research outputs.
- **Model File structure:**

- **Journal References:**

18.3 Code Repositories & Examples

Project	Repository	Description
Penobscot River	GitHub link	Simulates agent-based landward & seaward migration using STST.
Martha's Vineyard	GitHub link	Chloride cell regulation and stress-energy tradeoffs.

References

Glossary

This glossary defines key terms used throughout this book. Terms are organized alphabetically for clarity.

A

18.3.1 Acclimated Salinity

The salinity level a fish has gradually adjusted to and can tolerate without stress.

18.3.2 Agent

A virtual organism (e.g., fish) in the agent-based model that makes decisions and moves through space.

18.3.3 Allometric

A biological scaling relationship where physiological processes such as metabolism, growth, or filtration capacity scale nonlinearly with body size and/or age rather than proportionally (linearly).

18.3.4 Anadromous

Fish that hatch in freshwater, migrate to the ocean to grow, and return to freshwater to spawn.

18.3.5 Base Energy

The minimum energy required for basic physiological maintenance under resting conditions.

B

18.3.6 Benthic

Related to the bottom of a water body, where some fish feed or rest.

18.3.7 Bioaccumulation

The gradual buildup of contaminants like mercury in a fish over time.

18.3.8 Biomagnification

Increase in contaminant concentration as it moves up the food chain from prey to predator.

18.3.9 Bottom Feeding

A feeding strategy where fish forage along the riverbed or estuary floor.

18.3.10 Broadcast Spawning

A reproductive strategy where males and females release sperm and eggs freely into the water column.

C

18.3.11 Chloride Cells

Specialized gill cells that regulate salt balance during transitions between fresh and saltwater.

18.3.12 Contamination

Presence of toxins or pollutants (e.g., mercury) in water or sediment.

18.3.13 Current Velocity

Speed and direction of water movement at a location.

D

18.3.14 Diadromous

Fish that migrate between freshwater and saltwater as part of their life cycle.

18.3.15 Digestion

The physiological process by which ingested biomass is broken down and converted into usable energy.

18.3.16 Depth (Bathymetry)

Vertical distance from water surface to the bottom; influences habitat and hydrodynamics.

E**18.3.17 Ebb**

Outgoing tide when water flows toward the sea.

18.3.18 Environmental Stress

Physiological stress caused by environmental conditions such as salinity or temperature extremes.

F**18.3.19 Filter Feeding**

Feeding method where fish strain tiny particles (plankton, detritus, SPM) from water using gill structures.

18.3.20 Flood

Incoming tide when water flows inland.

18.3.21 Flood Limit

The maximum upstream reach of tidal influence.

H**18.3.22 Homing**

The ability of fish to return to their birthplace to spawn.

18.3.23 Homeostasis

Maintenance of stable internal conditions such as temperature and ion balance.

I**18.3.24 Iteroparous**

Species that can spawn multiple times across their lifespan.

L**18.3.25 Landward Migration**

Movement from the ocean into estuaries or rivers, often for spawning.

18.3.26 Lipid Catabolism

The mobilization and breakdown of stored lipids to meet energetic demands when food intake is insufficient.

M**18.3.27 Mercury**

A toxic metal that accumulates in aquatic systems and organisms.

18.3.28 Metabolism

The total energy expenditure required for physiological maintenance, swimming, digestion, osmoregulation, and other biological processes.

18.3.29 Methylmercury

A highly toxic, bioaccumulative form of mercury.

O

18.3.30 Optimal Temperature

Temperature range where fish perform best physiologically.

18.3.31 Osmoregulation

Physiological process of regulating internal salt and water balance.

18.3.32 Osmoregulatory Energy

Energy required to maintain ion balance under varying salinities.

P

18.3.33 Pairwise Spawning

A reproductive strategy requiring a male and female to occupy the same location for fertilization.

18.3.34 Patch

A discrete spatial cell in an agent-based model with its own environmental attributes.

18.3.35 Pelagic

Relating to open-water habitats away from the bottom.

S

18.3.36 Salinity

Concentration of dissolved salts in water.

18.3.37 Salinity Stress

Energy strain caused when salinity deviates from a fish's preferred range.

18.3.38 Salt Wedge

A dense layer of seawater intruding beneath freshwater in an estuary.

18.3.39 Seaward Migration

Movement from rivers or estuaries toward the ocean.

18.3.40 Selective Tidal Stream Transport (STST)

Behavior where fish exploit tidal currents to assist migration.

18.3.41 Semelparous

Species that spawn once and then die.

18.3.42 Spawning

Release of eggs and sperm for reproduction.

18.3.43 Staging

A temporary resting or holding phase in migration.

18.3.44 Suspended Particulate Matter (SPM)

Fine particles (sediment, organic matter) suspended in the water column.

18.3.45 Swimming Difficulty

Effort required to swim based on flow direction and strength.

18.3.46 Swimming Energy

Energy expenditure required for locomotion.

18.3.47 Swimming Speed (Velocity)

Rate of movement relative to water.

T**18.3.48 Temperature**

Water temperature; major driver of metabolism and behavior.

18.3.49 Temperature Stress

Stress caused when water is too warm or too cold for optimal performance.

18.3.50 Thermal Tolerance

Range of temperatures within which a fish can survive and function.

18.3.51 Thermoregulation

Behavioral regulation of temperature by moving into warmer or cooler habitats.